# Supplementary material: Complex mechanical learning with conservative KAM certification and dissipative magnetic feedback

The International Journal of Robotics Research
XX(X):1–40
©The Author(s) 0000
Reprints and permission:
sagepub.co.uk/journalsPermissions.nav
DOI: 10.1177/ToBeAssigned
www.sagepub.com/
SAGE

**Anonymous authors**

This supplement gives the full fixed-design learning, conservative KAM, power, sampled-feedback and numerical-interface arguments for the same frozen reduced model as the article. Appendix A supplies a sufficient nearby-design radius. The remaining appendices state the optimization and conditional statistical components separately from their experimental instantiation. The full-phase extension is analyzed separately and does not replace the article's certified structured Hamiltonian.

## S1 Mechanical model and complex structure

Let $q, p \in \mathbb{R}^2$ be configuration and canonical momentum, with normalized time $\tau = \omega_0 t$. All dots below denote derivatives with respect to normalized $\tau$. The reference is a specified finite-Gauss mechanical reduction, rather than an assertion of convergence to a three-dimensional rod PDE. On the configuration box $\Omega$, its known metric satisfies $G(q) = S(q)S(q)^T \succ 0$, where $S$ is lower triangular. Set

$$H_*(q,p) = \tfrac{1}{2} p^T G(q)^{-1} p + V_*(q), \quad V_*(q) = \tfrac{1}{2} q^T B q + 1 - \cos(d^T q), \quad B \succ 0. \tag{S1}$$

The magnetic port is $g(q) = d\cos(d^T q)$. A distributed linear-viscosity reference has $R_*(q) = a\{G(q) - K_r\} \succ 0$, for the fixed positive coefficient $a$ and rotary matrix $K_r$. Its coefficient is a declared numerical assumption, not an experimentally calibrated material law. Define the normalized reference damping matrix $M_* = S^{-1} R_* S^{-T} = a(I - S^{-1} K_r S^{-T})$, so $R_* = S M_* S^T$. Matrix 2-norms below are real Euclidean operator norms; the factor $a$ is already included in $M_*$. Writing $v = G^{-1} p$, the forced equations are

$$\dot{q} = v, \qquad \dot{p} = -\partial_q H_* - R_* v + g(q) u. \tag{S2}$$

Configuration-dependent inertia is retained: the $i$th inertial force is $v^T G_i v/2$, with $G_i = \partial_i G$. This is the canonical reduction appropriate to geometric mechanical models Antman (2005); Simo (1985).

The learned conservative energy is $\widehat{H} = p^T G^{-1} p/2 + \widehat{V}(q)$, with a trained real potential readout from constrained complex fully connected features. It is held fixed while a separate complex head learns a lower-triangular matrix $L_\theta(q)$. The latter has two literal complex tanh layers and a complex affine readout, followed by

$$\widehat{M} = \mathrm{Re}(L_\theta L_\theta^*), \qquad \widehat{R} = S \widehat{M} S^T, \qquad \widehat{\mathcal{D}}(q,v) = \tfrac{1}{2} v^T \widehat{R}(q) v. \tag{S3}$$

For real $z$, $z^T \widehat{M} z = \|L_\theta^* z\|_2^2 \geq 0$. Since the head depends only on $q$, $\partial_v \widehat{\mathcal{D}} = \widehat{R} v$. Thus the learned equations use $\widehat{H}$ in the canonical part and $-\partial_v \widehat{\mathcal{D}} + gu$ in the momentum equation. This follows the energy/port separation of HNN and dissipative models Greydanus et al. (2019); Zhong et al. (2020); Desai et al. (2021), with complex feature maps Trabelsi et al. (2018). The pair may be recorded as $\widehat{H} + i\widehat{\mathcal{D}}$; its dynamics are defined by this explicit decoder, not by assuming a holomorphic scalar generates arbitrary damping.

## S2 Weak observations and the optimization actually represented

### S2.1 Conservative training and distinct observation cohorts

The conservative potential was trained before the matrix dissipation head. Its known metric $G$, bending matrix $B$ and potential terms through degree four were prescribed. The full-cosine formula is available to the simulator and post-fit certification; the optimizer receives only trajectory observations. Its hidden higher-order remainder defines a controlled identification benchmark. Writing $s = d^T q$, the fitted potential is

$$\widehat{V}(q) = \tfrac{1}{2} q^T B q + \tfrac{1}{2} s^2 - \tfrac{1}{24} s^4 + \eta_V \sum_{a=0}^{6} q_1^a q_2^{6-a} c_a^{\mathrm{ev}}(q). \tag{S4}$$

Here $\eta_V = \mathrm{binary64}(10^{-5})$ is the stored correction scale. Each real coefficient function is the even symmetrization of a constrained complex fully connected network with two width-two literal complex-tanh hidden layers, evaluated with zero momentum coordinates. The seven networks supply 252 trainable real coordinates, including their real and imaginary parameter parts. This construction learns the sixth-and-higher potential correction while preserving the specified four-jet and inertia.

Table S1 separates the two COMSOL cohorts. Both cohorts use COMSOL Global Equations for the stated reduced chart. In the first stage, only the training trajectories' $q$, observed velocity, normalized time and prescribed input enter the optimization. A nonnegative scalar mass-proportional drag coefficient is profiled analytically at each objective evaluation; it is not supplied as a reference label. Initialization uses the same training weak equations. The 252 potential coordinates then undergo 400 accepted full-batch Armijo steps. The final weighted, scaled momentum-weak loss is

$$L_{\mathrm{mom}} = 1.6694817033398063 \times 10^{-10} > 0. \tag{S5}$$

The fixed kinematic term and parameter penalty are separate objective components. This recorded nonzero loss is not a claim of exact equation recovery or global optimization.

**Table S1.** Separate conservative and dissipative training cohorts. Counts distinguish training trajectories from reserved post-fit trajectories.

| Stage | Train/reserved | Normalized time | Trainable coordinates | Accepted steps |
|---|---|---|---|---|
| Conservative $\widehat{H}$ | $14/4$ | $[0, 0.2]$ | 252, scalar drag profiled | 400 |
| Dissipative $\widehat{M}$ | $14/4$ | $[0, 0.4]$ | 42, $\widehat{H}$ fixed | 100 |

The second stage uses a newly generated distributed-viscosity cohort. It keeps the trained $\widehat{H}$ fixed and fits the 42 real coordinates of the complex matrix head in (S3). A forty-coordinate linear potential nuisance is profiled in this stage and retained for error assessment; it is not added to the deployed $\widehat{H}$. The earlier scalar drag is not the distributed matrix model used in this stage. Subsequent observation-to-function and conservative KAM certificates concern the unchanged trained potential; new certificate data do not imply new conservative training. The complete procedure is sequential learning of a potential correction and a dissipative decoder, rather than simultaneous end-to-end identification of all mechanical parameters.

### S2.2 Weak matrix-head objective and optimization

For a recorded window, let $Q_w$ denote positive quadrature at its actual native times and let $\Pi(q, v) = G(q)v$. Define

$$a_i(V) = \tfrac{1}{2} v^T G_i v - \partial_i V + u g_i, \qquad [\mathcal{D}M]_{wi} = Q_w [S M S^T v]_i, \qquad r_Q = \mathcal{D}\widehat{M} - Q a(\widehat{V}) + \Delta\Pi. \tag{S6}$$

Only recorded $\tau, q, v, u$ are required; acceleration and reference damping labels are unnecessary. The dissipation fit fixes $\widehat{H}$ and trains the real coordinates $\theta$ of the damping head. Its cached numerical residual $r_f(\theta)$ includes row scaling, numerical quadrature and the fixed map $I - Q_f Q_f^T$. No exact orthogonality of $Q_f$ is assumed. Its real-arithmetic objective is

$$F(\theta) = \|r_f(\theta)\|_2^2 + \tfrac{\lambda}{2} \|\theta\|_2^2, \qquad \nabla F = 2 J^T r_f + \lambda\theta, \quad J = \partial_\theta r_f. \tag{S7}$$

The reported total adds a fixed kinematic constant, which leaves exact derivatives unchanged but can affect floating-point loss resolution. Strict learning certificates below reconstruct the native-time equations outwardly; they do not identify a rounded projected residual with an exact annihilation identity.

Complex coordinates $z = x + iy$ are optimized through their real and imaginary parts. For real $F$, $\partial_{\bar{z}} F = (F_x + iF_y)/2$ and $\mathrm{d}F[p_z] = 2\,\mathrm{Re}[(\partial_{\bar{z}} F)^* p_z]$. Holomorphicity of the loss is unnecessary. In the real-arithmetic lift, the smooth absolute bound is $\sqrt{b^2 + \epsilon_2}$ with fixed $\epsilon_2 > 0$. Constraint denominators are positive. Imaginary row budgets $\rho_1 = \rho_2 = 1/2 < \pi/4$ exclude the tanh poles: the first layer has $|\tanh| \le 1$, which supplies the second layer's component bounds. The readout and $\mathrm{Re}(LL^*)$ are real analytic. Cholesky acts on the fixed metric and is constant with respect to $\theta$. Hence $F$ is real analytic in its finite real coordinates.

**Proposition S1.** Exact-arithmetic idealization. *Fix a positive diagonal scale $D$, $\lambda > 0$, $\eta, \mu \ge 0$, $0 < c, \beta < 1$ and $\alpha_{\max} > 0$. The penalty includes every raw trainable real coordinate, and $r_f$ is real analytic as established above. An exact augmented least-squares step minimizes $2\|JDy + r_f\|^2 + \lambda\|Dy + \theta\|^2 + \eta\|Dy\|^2 + \mu\|y\|^2$. Its direction $p = Dy$ satisfies*

$$p = -B_o^{-1}\nabla F, \qquad B_o = 2J^T J + (\lambda + \eta)I + \mu D^{-2} \succ 0. \tag{S8}$$

*For $\nabla F \neq 0$, this and the fallback $p = -D^2\nabla F$ are strict descent directions. Unlimited exact Armijo backtracking takes the first successful geometric trial $\alpha_{\max}\beta^j$, $j = 0, 1, \ldots$, with $F(\theta + \alpha p) \le F(\theta) + c\alpha\nabla F^T p$, and terminates. An infinite exact iteration stops at stationarity or has $\|\nabla F(\theta_k)\| \to 0$.*

**Proof.** The normal equations give (S8); both directional derivatives are strictly negative. A local Lipschitz bound for $\nabla F$ gives Armijo for all sufficiently short steps. The initial $F$-sublevel is compact because $F \ge \lambda\|\theta\|^2/2$. There the preconditioners have common positive spectral bounds $m, M$, and all capped trial segments lie in a larger compact ball with a finite gradient Lipschitz bound $L$. Accepted steps are at least $\min\{\alpha_{\max}, 2\beta(1-c)m/(LM^2)\} > 0$. Summing Armijo decrease proves square-summability of gradient norms.

This adapts the descent component of (Zhang et al. 2022, Theorem 3) to the present objective; it does not transfer that paper's network, loss or update. The theorem for gradient iterations with exponential loss does not establish convergence of the present Gauss–Newton implementation. Rounded square roots, numerical rank truncation, least-squares error and rounded trial assignment remain outside the ideal proposition. The $10^{-6}$ gradient-consistency check is diagnostic. Stored records contain 2, 15 and 100 accepted steps for the constant, real and complex heads; the first two stop after 41 rejected trials. These records certify neither true-gradient error nor actual finite-precision convergence.

## S3 A deterministic observation-to-function bound

The finite design has fourteen training trajectories and 560 momentum-window components. Let $\chi$ contain the forty normalized monomials of total degrees 6, 8, 10 and 12, with normalization $r_0 = \mathrm{binary64}(0.1)$ interpreted exactly. The forty-five matrix coordinates contain all degree-at-most-four monomials in three symmetric entries. The common $G$, the prescribed low-order potential jet and the full-cosine reference are class assumptions justified independently of these finite observations. They are not conclusions of a finite-design rank calculation. Within this declared class, write

$$\widehat{M} = P_4(\widehat{c}) + \widehat{\rho}_M, \quad M_* = P_4(c_*) + \rho_{M,*}, \quad \widehat{V} - V_* = \chi^T a_V + \rho_V. \tag{S9}$$

These analysis representatives do not change the deployed functions. Set $N = Q\nabla\chi$, let $A$ be the matrix-polynomial weak design, and put $\widehat{t} = \mathcal{D}\widehat{\rho}_M$, $t_* = \mathcal{D}\rho_{M,*}$, $t_V = Q\nabla\rho_V$ and $e = Qa(V_*) - \mathcal{D}M_* - \Delta\Pi$. Subtraction of the finite equations gives

$$r_Q = A(\widehat{c} - c_*) + Na_V + \widehat{t} - t_* + t_V - e. \tag{S10}$$

The actual finite saved nuisance vector is denoted $b$, so $r_b = r_Q + Nb$; it is retained, not assumed to be exactly profiled away.

The observed $C = [N, A] \in \mathbb{R}^{560\times 85}$ has a certified left inverse. For fixed positive diagonal column scaling $D_c$, an approximate left inverse $L_c$ of $CD_c^{-1}$ is corrected using $T_c = L_c C D_c^{-1}$ and a strict bound $\|T_c - I\|_\infty < 1$. Consequently

$$K_V = [I_{40}, 0] D_c^{-1} T_c^{-1} L_c, \qquad K_V N = I_{40}, \quad K_V A = 0, \tag{S11}$$

and (S10) implies

$$a_V = \underbrace{K_V(r_b - \widehat{t}) - b}_{\text{observable principal term}} + K_V e + K_V t_* - K_V t_V. \tag{S12}$$

The subtraction of $b$ assesses the frozen deployed potential, not a virtual potential modified by nuisance coefficients. In particular, a small profiled residual controls the combined coordinates $a_V + b$, not $a_V$ alone: if the tails and consistency defect vanish and $\widehat{c} = c_*$, then $b = -a_V$ gives $r_b = 0$ even when $\chi^T a_V \neq 0$. The potential remains fixed during the dissipation fit, so the explicit $-b$ term in (S12) is required for its deployed error.

**Theorem S1.** Fixed-design conservative error. *On a real or complex domain $\mathcal{O}$, suppose, for a multi-index $\alpha$,*

$$B_\alpha \geq \sup_{\mathcal{O}} |(\partial^\alpha \chi)^T [K_V(r_b - \widehat{t}) - b]|, \quad \beta_\alpha \geq \sup_{\mathcal{O}} \left\|(\partial^\alpha \chi)^T K_V\right\|_2.$$

*If $\|e\|_2 \leq E$, $\|t_*\|_2 \leq T_*$, $\|t_V\|_2 \leq V_w$ and $\|\partial^\alpha \rho_V\|_{\infty,\mathcal{O}} \leq R_\alpha$, then*

$$\left\|\partial^\alpha(\widehat{H} - H_*)\right\|_{\infty,\mathcal{O}} \leq B_\alpha + \beta_\alpha(E + T_* + V_w) + R_\alpha. \tag{S13}$$

**Proof.** Multiply (S12) by $(\partial^\alpha \chi)^T$, apply Cauchy–Schwarz separately to its three remainder terms, and add $\partial^\alpha \rho_V$. The kinetic terms cancel identically.

For real boxes, signed Bernstein bounds control the principal polynomial and the squared output gain with Gram matrix $K_V K_V^T$. For complex rectangles, the finite Taylor identity for the degree-twelve polynomial transports all real derivative bounds, while analytic network and cosine tails are bounded independently. Thus real first-derivative control is not used as a substitute for complex-domain control.

Appendix A gives a local sufficient condition for fourteen prescribed trajectory directions, with an explicit initial-state radius and observation budget. It certifies observability of the finite representatives in (S9); the residual, nuisance and analytic remainders in (S13) remain required.

## S4 Conservative KAM certification for the same frozen family

The deterministic error bound addresses a fixed observed design and its independently justified consistency and class remainders. It supplies neither an iid sampling law nor an optimizer-success probability. In particular, the positive full-dimensional density and statistical risk assumptions used in the learning-to-KAM route of Chen et al. (2022) are not inferred from a finite set of trajectories. An iid design with positive full-dimensional density and a first-order Sobolev embedding with integrability exponent greater than four have not been instantiated for these two-degree-of-freedom data. The fixed-design route instead retains the explicit observable inverse and analytic remainder hypotheses.

For the frozen learned–reference pair, upward-rounded certified bounds are

| Configuration domain | $\lvert\widehat{H} - H_*\rvert$ | $\lvert\partial_1(\widehat{H} - H_*)\rvert$ | $\lvert\partial_2(\widehat{H} - H_*)\rvert$ |
|---|---|---|---|
| $\lvert q_i\rvert \leq 0.06$ | $5.910\,10^{-14}$ | $2.538\,10^{-12}$ | $1.989\,10^{-12}$ |
| $\lvert q_i\rvert \leq 0.1$ | $4.014\,10^{-12}$ | $4.583\,10^{-10}$ | $2.687\,10^{-10}$ |
| $\lvert\Re q_i\rvert \leq .06,\ \lvert\Im q_i\rvert \leq .01$ | $1.370\,10^{-13}$ | $8.325\,10^{-12}$ | $4.922\,10^{-12}$ |

The complex construction also bounds mixed derivatives through order three. A separately corrected damping-output inverse, with range $\mathbb{R}^{45}$, gives

$$\sup_{|q_i|\leq 1/10} \left\|\widehat{M} - M_*\right\|_2 \leq \delta_M < 2.719\,10^{-9}. \tag{S14}$$

Its principal term uses observations and learned functions; reference consistency, potential nuisance and both matrix tails remain separate. The damping inverse is not substituted for $K_V$.

Ordinary KAM theory concerns only the canonical fields of

$$H_\lambda = \widehat{H} + \lambda(H_* - \widehat{H}), \qquad 0 \le \lambda \le 1. \tag{S15}$$

Neither the viscosity nor the controlled input is part of this Hamiltonian family. Classical KAM requires genuine regularity, nonresonance and nondegeneracy hypotheses Pöschel (2001); positive mechanical stiffness alone is not an action-frequency twist condition.

For the present family, use period-one angles and the same exact Diophantine frequency $\nu$ throughout. The explicit approximate embeddings are

$$K_\lambda^{[3]} = K_0 + \lambda U_1 + \lambda^2 U_2 + \lambda^3 U_3, \qquad E_\lambda = X_{H_\lambda}(K_\lambda^{[3]}) - DK_\lambda^{[3]}\nu. \tag{S16}$$

For $\rho = 1/40$, the final residual norm is

$$\|E_\lambda\|_{\rho,\infty} := \max_i \sup_{|\operatorname{Im}\theta_j|<\rho} |(E_\lambda)_i(\theta)|. \tag{S17}$$

Low Fourier interpolants are bounded by weighted Fourier $\ell^1$ sums; the final bound in (S17) also includes aliasing and analytic tails and is not asserted to be a full weighted-$\ell^1$ norm. Complex analyticity, embedding rank, domain margins, mean-torsion inverses and all-integer Diophantine bounds $|k\cdot\nu| \ge \gamma_{\mathrm{D}}|k|_1^{-1}$ are verified together with the full hypotheses and scalar gates of (Haro and Luque 2019, Theorem 3.1). Here $\gamma_{\mathrm{D}}$ is a small-divisor constant, unrelated to viscosity. The corrected embeddings are analytic on $\rho_\infty = 1/400$.

The original accepted continuation route already consumes a continuous-data learning bound. It uses its own certified continuous reference IVPs and constant-damping identification class, rather than these native discrete observations. Its conservative discrepancy is reconstructed from continuous weak residuals, coefficient bounds and independent analytic tails, without a direct truth-gradient difference. That route remains valid for the same frozen conservative pair and frequency. The alternative native-source transport below rechecks the dependent geometric and smallness quantities. The auxiliary Fourier strip $\widehat{\rho} = 19/40$ uses an independent analytic proof on $|\Re(q,p)|_\infty, |\Im(q,p)|_\infty \le 9/20$; the smaller native error rectangle is not asserted to cover that domain.

### *S4.1 Native weak-observation derivative transport to KAM*

Write $D_V = \widehat{H} - H_*$, and let $b_{ab}$ denote the complete bound in (S13) for $\partial_1^a\partial_2^b D_V$ on $\mathcal{R} = \{|\Re q_i| \le 3/50, |\Im q_i| \le 1/100\}$. Because the kinetic terms agree, $D_V$ depends only on $q$. Its ordered derivative bounds give

$$\begin{aligned} \delta X_0 &= \max(b_{10}, b_{01}), & \delta X_1 &= \max(b_{20}+b_{11}, b_{11}+b_{02}),\\ \delta X_{t1} &= b_{20} + 2b_{11} + b_{02}, & \delta X_2 &= \max(b_{30}+2b_{21}+b_{12}, b_{21}+2b_{12}+b_{03}). \end{aligned} \tag{S18}$$

These respectively bound the component-maximum field error, induced matrix-infinity Jacobian error, total Hessian entry sum, and bilinear infinity-norm second field derivative. The last norm is $\max_i \sum_{j,k} \sup|T_{ijk}|$; ordered pairs $(1,2)$ and $(2,1)$ both contribute, explaining the factor two. The $t1$ convention is a total Hessian entry sum, not an induced norm of a transposed Jacobian.

The image of $K_0$ on the target strip, enlarged by radius $61/20000$, and the retained derivative discs lie in $\mathcal{R}$. The same cubic response obeys $\sum_{j=1}^3 \|U_j\|_\rho + 3/1000 < 61/20000$. Adding (S18) to the known learned-field bounds therefore controls the family on this common tube. At fixed $K_0$, its canonical normal frame $N_0$ gives

$$T_\lambda - T_0 = -\lambda N_0^T \nabla^2(H_* - \widehat{H})(K_0)N_0, \qquad \|\langle T_\lambda - T_0\rangle\|_\infty \le \left\|N_0^T\right\|_\infty \delta X_1 \|N_0\|_\infty . \tag{S19}$$

The corresponding Neumann test remains strict. Repeating the varying-map Gram and mean-torsion perturbation calculation with these new bounds also preserves nondegeneracy for the same $K_\lambda^{[3]}$; its margins and Haro inputs are recomputed, rather than copied from the previous learning source.

On the target-strip product domain with $|\lambda| \le R = 10^5$, the accepted response-support bound keeps $K_\lambda^{[3]}$ within the retained $3/1000$ tube also on this complex parameter disk. Let $M_{\widehat{X}}$ bound the learned field and set $\mathcal{T} = \sum_{j=0}^{3} R^j \left\| D_\nu U_j \right\|_\rho$, $U_0 = K_0$. The new residual modulus and parameter tail are

$$M_A = M_{\widehat{X}} + 10^5 \delta X_0 + \mathcal{T}, \qquad E_{\mathrm{par}} = \frac{M_A}{R^8 (R-1)}. \tag{S20}$$

The accepted low Fourier bounds for coefficients $e_0, \dots, e_8$ are shared because the Hamiltonian family, frequency and cubic response are identical. The independent larger-domain analytic bound supplies the Fourier alias/analytic tail $E_\theta$, while (S20) supplies a fresh parameter tail. With $I_j$ the corresponding low-interpolant component bounds, the recomposed estimate is

$$\left\| E_\lambda \right\|_{\rho,\infty} \le \sum_{j=0}^{8} I_j + E_\theta + E_{\mathrm{par}} < 1.205897\, 10^{-41}. \tag{S21}$$

The native model discrepancy and the embedding invariance defect are different quantities. The transported first-derivative estimate $\delta X_0 < 8.324054\, 10^{-12}$ bounds $X_{H_*} - X_{\widehat{H}}$ on the configuration rectangle; it cannot replace the residual $E_\lambda$ in (S17), which is evaluated along the corrected embedding on the full angular strip. The cubic response correction in (S16) and the complete Fourier and parameter-tail estimates produce the much smaller invariance defect required by the a-posteriori theorem. All eight canonical numerical gates hold with the transported native bounds; the fresh sufficient threshold exceeds $5.82604\, 10^{-38}$. The a-posteriori theorem therefore gives a real analytic embedded invariant two-torus on $\rho_\infty = 1/400$ at the fixed frequency for each real $\lambda \in [0, 1]$. The native weak observations now enter this KAM route through both derivative/nondegeneracy transport and the recomputed parameter tail. The prior continuous-data route remains a separate valid argument. In neither route does a function-error bound alone verify the embedding residual. The conclusion asserts neither a mixed model-error ball, analytic dependence of exact tori on $\lambda$, a phase-space percentage, nor a necessary threshold for chaos.

## S5 Mechanical power and continuous magnetic stabilization

The real energy derivative of the learned dissipative system is exactly

$$\dot{\widehat{H}} = -v^T \widehat{R} v + uy, \qquad y = g(q)^T v. \tag{S22}$$

Indeed, the canonical contributions cancel by skew symmetry, and (S3) supplies the dissipated power. If the reference is driven by the same input, shared inertia and the bounds $|\partial_i(\widehat{V} - V_*)| \le \varepsilon_i$ give

$$\dot{\widehat{H}} \le -v^T \widehat{R} v + uy + \sum_i \varepsilon_i |v_i| + \delta_M v^T G v. \tag{S23}$$

The matrix-error contribution follows by setting $z = S^T v$ in $|z^T(\widehat{M} - M_*)z| \le \delta_M \left\| z \right\|^2$. If $M_* \succeq m_* I$ and $\delta_M < m_*$, then

$$\widehat{M} \succeq (m_* - \delta_M) I, \qquad v^T R_* v \ge \frac{m_*}{m_* + \delta_M} v^T \widehat{R} v. \tag{S24}$$

Identification accuracy thus controls usable physical dissipation credit; the PSD property itself already holds algebraically.

For continuous stabilization of (S2), put $s = d^T q$ and use the frozen learned-equilibrium feedforward $T = 6380576977437163/144115188075855872$. With $s_* = 3/1000$, $k_p = 4$, $k_d = 3$, $a_p = 1/50$ and $a_d = 3/100$, define

$$u_b = \frac{T - \mathrm{sat}_{a_p}(k_p(s - s_*)) - \mathrm{sat}_{a_d}(k_d \dot{s})}{\cos s}, \qquad \Psi(e) = \int_0^e \mathrm{sat}_{a_p}(k_p r)\, \mathrm{d}r. \tag{S25}$$

The shaped storage $J_b = H_* - Ts + \Psi(s - s_*)$ obeys $\dot{J}_b = -v^T R_* v - \dot{s}\,\mathrm{sat}_{a_d}(k_d\dot{s}) \le 0$. On the certified configuration box $\cos s \ge c_b > 0$ and $D^2 V_* \succeq B \succ 0$; $\Psi$ is convex and $C^1$ with locally Lipschitz derivative. A certified interior root gives its unique equilibrium. In its inactive-saturation neighborhood, writing $\beta_d = d^T B^{-1} d$, this point satisfies

$$(\beta_d^{-1} + k_p)s_e + \sin s_e = T + k_p s_*, \qquad q_e = B^{-1} d\, s_e/\beta_d, \quad p_e = 0. \tag{S26}$$

For completeness, let $c$ be an energy cap excluding configuration faces, let $S_c$ bound $|s|$ on $H_* \le c$, and choose $\overline{J} \ge J_b(x_0)$ such that $\overline{J} + |T|S_c < c$. Before any first cap exit, $H_* = J_b + Ts - \Psi \le \overline{J} + |T|S_c < c$. This contradiction excludes the exit; configuration faces are excluded by their energy barrier. The resulting component is compact and forward invariant. Its largest invariant zero-dissipation subset has $v = p = 0$ and then the unique $q_e$, proving convergence by the invariance argument. No twice differentiable saturation is required.

For the declared buffers, such a component contains $q_0 = (1/80, 1/80)$, $p_0 = (1/200, 1/1000)$. It respects $|u_b| < 0.094416 < 0.1$ and has $|s_e - s_*| < 1.334\,10^{-19}$, including feedforward-buffer rounding. These are ideal continuous-reference conclusions. Under a held command, $\dot{J}_b$ gains $(u_n - u_b(x))\cos s\,\dot{s}$; its sign is not controlled merely by an energy cap. Sampled stabilization therefore requires a separate argument.

## S6 Local sampled contraction and bounded-error practical stability

Let $x = (q, p)$, let $\pi(x)$ be (S25), and let $\Phi_h(x, u)$ denote the exact reference flow with a constant input. The ideal sample map is $\mathcal{S}_h(x) = \Phi_h(x, \pi(x))$. Its equilibrium linearization uses the fixed-command plant derivative, including the nonzero equilibrium input:

$$\begin{aligned} A_p &= \begin{pmatrix} 0 & G_e^{-1} \\ -B - (\cos s_e + u_e \sin s_e)dd^T & -R_e G_e^{-1} \end{pmatrix}, \quad B_p = \begin{pmatrix} 0 \\ d\cos s_e \end{pmatrix}, \\ D\mathcal{S}_h(x_e) &= e^{A_p h} + \int_0^h e^{A_p t} B_p \,\mathrm{d}t\, D\pi(x_e). \end{aligned} \tag{S27}$$

This is the zero-order-held map, not the exponential of the continuous closed-loop Jacobian.

On the single normalized clock window $h \in [0.1 - 10^{-12}, 0.1 + 10^{-12}]$, a fixed dyadic matrix $P_L$ satisfies $4I \prec P_L \preceq 74.237I$. Set

$$V_L(x) = (x - x_e)^T P_L (x - x_e), \qquad \mathcal{E} = \{V_L \le 81\,10^{-6}\}, \qquad R = 9/1000. \tag{S28}$$

The ellipsoid is contained in the radius-$0.0045$ Euclidean ball about $x_e$ and hence its enclosing component cube. On that cube the two saturations are inactive. A complete held-flow Jacobian enclosure and a nonnegative augmented comparison exponential prevent first exit from the radius-$0.015$ outer component box during each hold. Duhamel transport of the variational equation gives a uniform quadratic derivative decrement. Integrating the derivative along the segment from $x_e$ and using convexity of the quadratic form yields

$$V_L(\mathcal{S}_h(x)) \le 0.987115\, V_L(x), \qquad x \in \mathcal{E}. \tag{S29}$$

This establishes local ideal sample convergence for arbitrary clock variation in the one stated window. On the verified outer box the held flow and $\pi$ are locally Lipschitz, the holds are uniformly bounded, and $\Phi_t(x_e, \pi(x_e)) = x_e$. A uniform bound $\sup_{0\le t\le h} \|\Phi_t(x, \pi(x)) - x_e\| \le C\,\|x - x_e\|$ therefore transports sample convergence to uniform intersample convergence. It does not place the original experiment initial state in this local basin.

**Theorem S2.** Conditional practical sampled stability. *Suppose the true initial state lies in $\mathcal{E}$, every hold is in the stated window, and the actual held command is $u_n = \pi(x_n) + \delta_n$ with $|\delta_n| \le \epsilon_u = 10^{-5}$. The enlarged held-flow first-exit bounds and input-sensitivity calculation give*

$$r_{n+1} \le c_s r_n + L_u \epsilon_u, \qquad r_n = \sqrt{V_L(x_n)}, \quad c_s = 497/500, \quad L_u = 224417/10^6. \tag{S30}$$

*Consequently $\mathcal{E}$ is sample-invariant, and* $\limsup r_n \le L_u\epsilon_u/(1 - c_s)$.

**Proof.** Interpolating between the ideal and perturbed held inputs within the verified outer box gives

$$\|\Phi_h(x,\pi(x)+\delta)-\mathcal{S}_h(x)\|_{P_L}\le L_u|\delta|.$$

The triangle inequality and $\sqrt{0.987115}<c_s$ prove (S30). Since $c_sR+L_u\epsilon_u\le R$, induction proves invariance; summing the geometric recurrence gives the limit bound.

For $d_4=(d_1,d_2,0,0)^T$ and $D_P=\sqrt{d_4^TP_L^{-1}d_4}$, $\limsup|d^Tq_n-s_*|\le D_PL_u\epsilon_u/(1-c_s)+|s_e-s_*|<4.904\,10^{-5}$. Observation, numerical-evaluation and safety-filter errors must jointly establish the command premise. Persistent errors yield this practical sample-time bound, not exact convergence. Neither finite point checks nor an alternative fixed-envelope filter proves the actual algorithm's uniform future behavior.

## S7 Composition with an executed finite record

Let $x_n$ be the true reference state and $y_n$ an actually recorded native observation. A continuous IVP enclosure must cover the same executed held inputs and transport its error through every slab and hold, without resetting uncertainty at observations. Suppose it establishes $|y_{n,i}-x_{n,i}|\le\rho_{n,i}\le10^{-7}$ and endpoint boxes. Actual normalized hold lengths are computed from exact stored SI times and $\omega_0$; they must satisfy the window in Section S6. The output $d^Tq$ is a generalized angle, not a displacement in millimetres.

Let $u_n^{\mathrm{num}}$ be the stored numerical request and $u_n$ the exactly read-back executed command. Independent pointwise arithmetic and transcript verification must provide

$$|u_n^{\mathrm{num}}-\pi(y_n)|\le e_{\mathrm{eval},n},\qquad |u_n-u_n^{\mathrm{num}}|=e_{\mathrm{gate},n},\qquad \eta_n=\sum_i K_i\rho_{n,i}+e_{\mathrm{eval},n}+e_{\mathrm{gate},n},\tag{S31}$$

where $K_i$ bounds $|\partial_i\pi|$ on the verified local cube. Its geometry includes the prescribed observation boxes of every true state in $\mathcal{E}$.

**Theorem S3.** Finite-tail composition. *Suppose a complete independently validated record of $N$ holds has a true endpoint box at index $j$ contained in $\mathcal{E}$. For every already executed $j\le n<N$, assume the clock window and $\eta_n\le\epsilon_u$ are verified. For any outward $\overline{r}_j\ge\sqrt{V_L(x_j)}$ with $\overline{r}_j\le R$, define*

$$\overline{r}_{n+1}=c_s\overline{r}_n+L_u\eta_n.\tag{S32}$$

*Then $x_n\in\mathcal{E}$ and $\sqrt{V_L(x_n)}\le\overline{r}_n\le R$ for $j\le n\le N$.*

**Proof.** The observation geometry and mean-value theorem give $|\pi(y_n)-\pi(x_n)|\le\sum_i K_i\rho_{n,i}$. Thus (S31) bounds the true held-input discrepancy. The robust held-flow certificate gives (S32); $c_sR+L_u\epsilon_u\le R$ closes finite induction.

Entry can be checked without replacing the true state by the numerical center: subtract the equilibrium box from the true endpoint box and maximize $e^TP_Le$ at its sixteen vertices. Convexity in each coordinate makes this a valid whole-box upper bound. Its first successful index is a first *guaranteed* entry, not necessarily the true first hitting time. The finite theorem does not certify commands beyond $N$, and a partial record does not instantiate it for a declared full horizon.

The power and learning-domain statements compose separately. The executed gate retains its original full-0.1-box budgets, $\delta_M^{\mathrm{run}}\simeq2.8220300339\,10^{-9}$ and $\varepsilon_0^{\mathrm{run}}\simeq2.7717816565\,10^{-13}$, rather than silently adopting the new observation bounds. The new damping bound is smaller on the whole 0.1 box. The new conservative energy and gradient bounds dominate the older numerical budgets only on the 0.06 box; the new 0.1-box bounds do not provide that dominance.

A ledger must bound physical energy throughout each hold, not just at its endpoint. Its original full-0.1 validity must first be established independently, including the original cap/domain and reference-IVP requirements. Using that valid ledger, rather than assuming the desired smaller domain to justify it, suppose its within-hold bound $W_n<c_r=10^{-3}$ holds for every relevant hold and $q$ starts in $|q_i|<r=0.06$, then

$$E_{\mathrm{face}}=\min_i\frac{r^2}{2(B^{-1})_{ii}}>c_r\quad\Longrightarrow\quad|q_i|<r.\tag{S33}$$

Indeed $1-\cos(d^T q)\geq 0$, and minimizing the elastic quadratic at a configuration face gives $E_{\mathrm{face}}$, excluding a first exit. Only after this independent first-exit argument do the smaller-domain learning bounds and (S23) apply to the executed trajectory. Thus no local-error/ledger bootstrap is used circularly. A controller using a larger energy cap does not automatically satisfy this smaller-ledger condition. The executed gate retains its own frozen error budgets; using tighter local identification bounds in (S23) is a separate posterior composition, not a change to that executed algorithm.

The resulting framework distinguishes conservative invariant-torus existence, physical dissipated power, continuous stabilization, conditional sampled practical stability and a validated finite executed tail. Continuous final-hold tracking requires its own intersample tube bound. Unknown-plant calibration, velocity-dependent Morison drag, contact and continuous-rod error require additional modeling evidence. Statistical generalization and complex-over-real advantage are not consequences of the stated deterministic results.

## S8 A continuing numerical decision rule and its finite antecedent

Theorem S3 applies to commands that have already been executed. Continuing control requires a rule whose numerical output and energy-ledger update satisfy uniform conditions at every subsequent sample. The construction below uses the same physical reference, conservative potential, dissipative head and exact ideal law $\pi$ from (S25). It is a separate numerical implementation; its binary64 outputs need not coincide with those of the finite experiment. The local held-flow analysis of Section S6 is retained.

### *S8.1 Observation domains and uniform numerical returns*

Write $\rho_{\mathrm{obs}}=10^{-7}$ and let $\mathcal{X}_{\mathrm{obs}}$ be the fixed rational rectangle containing $\mathcal{E}\oplus[-\rho_{\mathrm{obs}},\rho_{\mathrm{obs}}]^4$. The sharp ellipsoid-coordinate bounds place every admissible observed center $z$ in $\mathcal{X}_{\mathrm{obs}}$. The entire radius-$\rho_{\mathrm{obs}}$ evaluator box about such a center lies in the initial derivative cube and has $|q_i|<3/50$. The evaluator box is not required to be contained in $\mathcal{X}_{\mathrm{obs}}$ itself. Nor is the initial derivative cube assumed to contain every point of a held trajectory; that trajectory uses the separate outer domain of the robust held-flow theorem.

The same nominal derivative bounds used in (S31) give

$$|\pi(z)-\pi(x)|\leq e_{\mathrm{obs}}=\frac{295416611}{335544320000000},\qquad x\in\mathcal{E},\quad \|z-x\|_\infty\leq\rho_{\mathrm{obs}}. \tag{S34}$$

All statements about a true state below require this inclusion condition; normal return at an arbitrary software input does not establish it.

The numerical nominal evaluator uses rational coefficient enclosures and analytic tails for a degree-ten Taylor representation of the known metric. At a point query it evaluates monomials exactly and forms the two-by-two inverse by the adjugate/determinant formula. On $\mathcal{X}_{\mathrm{obs}}$, let $B_G$ bound all metric-entry magnitudes, $d_0>0$ bound the determinant from below, $e_G$ bound each point-query entry radius and $p_{\max}=\sup\|p\|_\infty$. Interval diameter estimates give

$$r_{\det}\leq 4B_G e_G,\qquad r_v\leq\frac{2p_{\max}e_G}{d_0}+\frac{2p_{\max}B_G r_{\det}}{d_0^2}. \tag{S35}$$

These are point-query radii uniformly over the rectangle, rather than the diameter of the whole-domain function image. With $s=d^Tq$, cosine is enclosed by the exact even polynomial through degree twelve and the Lagrange remainder $|s|^{14}/14!$. If $S=\sup_{\mathcal{X}_{\mathrm{obs}}}|s|$ and $r_c=S^{14}/14!$, its lower endpoint is at least $c_0=1-S^2/2-2r_c>0.99$. Exact endpoint clipping implements the saturations in (S25). Consequently the nominal interval radius is bounded by

$$r_\pi\leq\frac{k_d\,\|d\|_1\,r_v}{c_0}+\frac{(|T|+a_p+a_d)r_c}{c_0^2}. \tag{S36}$$

Let $g_E=2^{-80}$. The implementation combines a 256-bit learned-energy enclosure with the 192-bit sampled power and dissipation-credit enclosure. Their fixed cache construction is retained; each energy query sets its 256-bit context, and the sampled context is restored to 192 bits before the sample query. The dissipative and conservative budgets remain $\delta_M^{\mathrm{run}}$ and $\varepsilon_0^{\mathrm{run}}$ from Section S7. They are not replaced by later, smaller identification bounds during the numerical update.

The computational premises are correspondence of the specified interval library with its enclosure semantics, valid outward elementary operations, exact integer/rational arithmetic, ordinary binary64 basic arithmetic and adequate resources. Calls use the fixed caches, are serialized, and admit no external mutation of the process-global interval precision. Observed coordinates are finite binary64 numbers. Ledger inputs are bounded grid rationals in their canonical finite representation. Outward conversion of the requested radius is included in the evaluator bounds; exact-zero shortcuts, pole exclusions and all computed energy derivatives belong to the successful-return assertion.

**Proposition S2.** Uniform numerical decision interfaces. *Under these computational premises, every admissible observed center $z$ has a normally returning numerical nominal request $u^b = j2^{-40}$ with*

$$|u^b - \pi(z)| \le \delta_u < 4.547473509315\,10^{-13}, \qquad |u^b| < 0.094275056129 < 1/10. \tag{S37}$$

*The energy and sample evaluators return finite quantities satisfying*

$$\begin{aligned} E^+ &< 0.000338941808 < 1/1000, \qquad E^+ \in g_E\mathbb{Z}_{\ge 0}, \\ |\widetilde{y}| + e_y &\le 0.009921509458 < 1/50, \qquad e_y \ge 0, \quad \underline{d} \ge 0. \end{aligned} \tag{S38}$$

*If the true sample $x$ lies in the observed box, their enclosures also give*

$$H_*(x) \le E^+, \qquad |y(x) - \widetilde{y}| \le e_y, \qquad v(x)^T R_*(q(x))v(x) \ge \underline{d}. \tag{S39}$$

**Proof.** The metric and cosine enclosures contain their exact values and have the strict denominator margins stated above. Rounding the nominal interval midpoint to the nearest $2^{-40}$ grid point gives error at most $r_\pi + 2^{-41}$. The uniform bound $|j| < 2^{37}$ makes conversion of $j2^{-40}$ to binary64 exact. Diagnostic endpoints are rounded outward on $g_E$, and their error bound is rounded upward on that grid, adding at most $2g_E$. This gives (S37) without altering the selected command. Every reported nominal fraction has at most 52 characters.

The actual energy and sampled evaluator graphs are bounded uniformly over all admitted center ranges and outward input-ball radii. Composition of their elementary-operation and complex-network bounds proves the finite returns and strict guards in (S38), including nonnegative sampled error and credit. Every internally formatted energy integer has at most 520 decimal digits. The fresh candidate is the upward grid rounding of the nonnegative part of the learned-energy upper endpoint plus $\varepsilon_0^{\mathrm{run}}$; outward rounding preserves its physical upper-bound property. The conservative error certificate and damping credit construction therefore yield (S39). These are bounds on actual numerical returns as well as on ideal function values.

## S8.2 Joint state–ledger invariance

The fixed runtime cap is

$$c_{\mathrm{rt}} = \frac{5955826498595333236954919797339283}{25124747867899983792704726160008806 40} \simeq 0.002370501997. \tag{S40}$$

It is distinct from both the smaller recorded-energy threshold $c_r$ in Section S7 and the fresh-candidate bound $R_E = 1/1000$. Given a prediction $W_n$, the rule first sets $W_n^{\mathrm{reset}} = \min(W_n, E_n^+)$ and then applies the same exact scalar power gate with hold budget $h_+ = 1/10 + 10^{-12}$, actuator bound $U = 1/10$ and physical derivative bounds $L_y, L_d$. Its candidate satisfies the power inequality

$$h_+(u\widetilde{y} + |u|e_y - \underline{d}) + \tfrac{1}{2}h_+^2(|u|L_y + L_d) \le \tfrac{1}{2}(c_{\mathrm{rt}} - W_n^{\mathrm{reset}}). \tag{S41}$$

The next ledger rounds the nonnegative endpoint prediction upward on $g_E$ and takes its maximum with $W_n^{\mathrm{reset}}$; the zero-input branch preserves that reset ledger. The maximum supplies a bound for every actual duration up to the hold budget.

**Theorem S4.** Continuing numerical state–ledger invariance. *Suppose $x_0 \in \mathcal{E}$ and $H_*(x_0) \le W_0 < c_{\mathrm{rt}}$, with $W_0 \in g_E\mathbb{Z}$. At every subsequent sample suppose that a finite binary64 observation obeys $\|z_n - x_n\|_\infty \le \rho_{\mathrm{obs}}$, the returned binary64 command is executed exactly and constantly over the entire actual hold, and*

$$h_n \in [h_-, h_+], \qquad h_\pm = 1/10 \pm 10^{-12}. \tag{S42}$$

*Assume the same reference* (S2)*, with its declared viscosity and no additional external force, and the computational premises of Proposition S2. Every decision invocation then returns normally, preserves the numerical nominal request through the gate, and maintains*

$$x_n \in \mathcal{E}, \qquad H_*(x_n) \le W_n < c_{\mathrm{rt}}, \qquad W_n \in g_E\mathbb{Z}. \tag{S43}$$

*On each entire hold,* $H_* < 0.001441122076$ *and* $|q_i| < 69/1250$.

**Proof.** Assume (S43) at a sample. The observed center and its entire evaluator box belong to their respective certified domains. Both reset operands bound the same physical energy, so their minimum is valid, lies on $g_E\mathbb{Z}$ and is at most $R_E$. By (S38), every $u \in [-U, U]$ has increment at most

$$I = h_+U/50 + \tfrac{1}{2}h_+^2(UL_y + L_d) < \tfrac{1}{2}(c_{\mathrm{rt}} - R_E). \tag{S44}$$

Hence the entire actuator interval satisfies (S41). The request $u_n^b$ is an exactly representable feasible candidate at zero distance from itself and is therefore preserved. The strict grid margin gives

$$W_{n+1} < B_E := R_E + I + g_E < 0.001441122076 < c_{\mathrm{rt}}, \qquad W_{n+1} \in g_E\mathbb{Z}, \tag{S45}$$

including the zero-input branch and the maximum with the reset ledger.

The mean-value estimate and numerical error bound yield

$$|u_n^b - \pi(x_n)| \le e_{\mathrm{obs}} + \delta_u < 8.804106\,10^{-7} < \epsilon_u. \tag{S46}$$

Theorem S2 therefore supplies the complete canonical held-flow domain, recurrence (S30) and $x_{n+1} \in \mathcal{E}$.

To close the energy part, apply the physical whole-hold power inequality up to its first energy or configuration exit. Its quadratic-in-time upper bound is at most the larger of the reset and rounded endpoint ledgers. With $E_g$ the physical derivative guard, the strict comparisons are

$$B_E < E_{\mathrm{face}} = \min_i \frac{(3/50)^2}{2(B^{-1})_{ii}}, \qquad B_E < c_{\mathrm{rt}} < E_g.$$

They exclude those stopped exits. The canonical state domain follows separately from the held-flow theorem. The additional comparison $(69/1250)^2 > (3/50)^2 B_E/E_{\mathrm{face}}$ yields the stated configuration bound from $H_* \ge q_i^2/[2(B^{-1})_{ii}]$. Thus the physical endpoint energy is bounded by $W_{n+1}$ and the next observation boxes remain in the learning domain. The reset, gate and final maximum preserve the grid. The uniform finite-return and bounded-formatting properties give normal return at each invocation, completing joint induction. No numerical IVP uncertainty is reset in this argument.

### *S8.3 Continuous practical tracking and terminal initialization*

A continuous tracking bound requires transport through the entire hold, rather than applying a sample-time output estimate between samples. Let $F_*(x, u)$ denote the reference state field and let $C \ge 0$ be its certified componentwise absolute Jacobian majorant for the five-dimensional augmented field $(F_*(x, u), 0)$ on the shared outer domain. Put $\mathsf{T}_h \ge \exp(Ch_+)$ componentwise, $w_j \ge \sqrt{(P_L^{-1})_{jj}}$ and $a_\pi = \sum_{j=1}^4 K_j w_j$. Then the initial augmented error is bounded by $(w_1 r_n, \ldots, w_4 r_n, a_\pi r_n + \epsilon_u)$. Stopped-flow comparison on the same outer domain gives

$$\sup_{0 \le t \le h_n} |d^T q(\tau_n + t) - s_*| \le C_r r_n + C_\epsilon \epsilon_u + \beta_e, \tag{S47}$$

where $\beta_e \geq |d^T q_e - s_*|$ and

$$C_r = \sum_{i=1}^{2} |d_i| \left( \sum_{j=1}^{4} (\mathsf{T}_h)_{ij} w_j + (\mathsf{T}_h)_{i5} a_\pi \right),$$
$$C_\epsilon = \sum_{i=1}^{2} |d_i| (\mathsf{T}_h)_{i5}. \tag{S48}$$

The nonnegative matrix exponential is bounded by its rational Taylor sum through order 64 and the entrywise tail $M^{65}/[65!(1 - M/66)]$, where $M = \|Ch_+\|_\infty < 66$. The resulting constants satisfy $C_r < 0.286233270218215$ and $C_\epsilon < 0.003548859583130$. The output in (S47) is the generalized angle $d^T q$, not the power port $y$.

**Corollary S1.** Continuing entire-hold tracking. *Under Theorem S4, for any continuing sequence of the stated holds,*

$$r_n \leq c_s^n R + (1 - c_s^n) r_\infty, \qquad r_\infty = \frac{L_u \epsilon_u}{1 - c_s} = \frac{224417}{600000000}, \tag{S49}$$

*and*

$$\limsup_{\tau \to \infty} |d^T q(\tau) - s_*| < 0.000107094841601. \tag{S50}$$

*The entire hold at index 306 and every later hold satisfy $|d^T q - s_*| < 1/2000$, when index zero begins at any initialization meeting the joint state–ledger hypotheses.*

**Proof.** Sum (S30) to obtain (S49) and insert it in (S47). Since $h_- > 0$, sampling times cannot accumulate and the sample index tends to infinity with normalized time. Taking the upper limit gives (S50). The exact sufficient output majorant decreases from a value above $1/2000$ at index 305 to a value below it at index 306. This establishes a sufficient bound index, not a first hitting time or an executed duration.

The validated finite record of Section S7 contains 200 holds and 20,000 enclosed integration slabs. Its first guaranteed whole-tube entry is at sample 199, leaving one actually executed post-entry hold. At sample 200 the complete terminal state tube satisfies

$$\sup_{x \text{ in terminal tube}} V_L(x) \leq 7.816222\,10^{-5} < 81\,10^{-6},$$
$$W_{\text{term}} = \frac{14010608820076932197}{75557863725914323419136} \in g_E \mathbb{Z}, \qquad W_{\text{term}} < c_{\text{rt}}. \tag{S51}$$

The whole-tube quadratic bound includes the equilibrium uncertainty and all sixteen vertices. The physical IVP and power proof supplies $H_*(x_{200}) \leq W_{\text{term}}$; the ellipsoid check alone does not imply that energy-ledger statement. Every admissible new observation of this terminal tube belongs to $\mathcal{X}_{\text{obs}}$. These facts supply a valid initialization for Theorem S4.

The prospective decision program was subsequently connected to COMSOL for two further holds, initialized by the sample-200 native state, the carried ledger in (S51) and the positive reference-IVP error radius. Each command was computed from the preceding native endpoint by the specified decision class, rather than supplied as a command replay. The executed commands were

$$u_{200} = \frac{49692742889}{2^{40}}, \qquad u_{201} = \frac{49763151619}{2^{40}}. \tag{S52}$$

Both complete native-state restarts and command readbacks agreed bitwise. Independent validation covered all $2 \times 100$ new integration slabs, with the initial positive error radius inherited from the preceding 20,000-slab proof and never reset. The coordinate error at either new sample endpoint was less than $3.524\,10^{-13}$. The actual raw-SI hold durations satisfied (S42); the command, ledger and actual-time chains closed at sample 202 with

$$W_{202} = \frac{1109624721490629148 33}{60446290980731458735 3088} < c_{\text{rt}}. \tag{S53}$$

At sample 202, a sixteen-vertex bound including equilibrium uncertainty gives $\sup V_L \leq 7.540\,10^{-5} < 81\,10^{-6}$ on the entire enclosed terminal state tube. Together with its validated energy ledger, this retains the initialization conditions for continuing control. Thus the finite evidence comprises 200 holds under the preceding program and two verified holds

under the prospective program. This switch does not constitute an executed 306-hold record for the latter, nor replace the preceding record's independent final-50-hold continuous tracking result. The continuing theorem applies to subsequent holds under its stated conditions. Persistent bounded errors yield practical bounds; they do not imply zero tracking error. Statistical generalization, complex-over-real superiority, hardware calibration, a dissipative KAM theorem and continuum-rod approximation remain outside these conclusions. Finite normal return does not establish a real-time deadline or bounds on sensing, computation and actuation latency; the control conclusions retain their stated actual-input and hold-clock assumptions.

## A Robustness of a fourteen-direction data design

This appendix gives a deterministic sufficient condition for nearby complete bundles of fourteen trajectories of the declared reduced reference. The nominal observation matrix, centre-trajectory enclosures and feature bounds supply a positive certified initial-condition radius. The argument retains the quadrature, observation errors, finite-class remainders and nuisance coordinates; independent sampling in phase space is not assumed.

*The actual finite rows and their signs.* Let $r_0$ be the exact stored binary64 value of $0.1$ and write

$$\chi_{ab}(q) = (q_1/r_0)^a (q_2/r_0)^b, \quad a+b \in \{6,8,10,12\}, \qquad \psi_{ab}(q) = (q_1/r_0)^a (q_2/r_0)^b, \quad a+b \le 4.$$

Their orders are increasing total degree and then increasing $a$. There are forty $\chi$ and fifteen $\psi$. Set $E_1 = E_{11}$, $E_2 = E_{12} + E_{21}$, $E_3 = E_{22}$ and $G(q) = S(q)S(q)^{\mathsf{T}} > 0$, using the lower Cholesky factor. For each trajectory there are twenty windows of forty intervals, with two momentum components per window. The exact observed nodes and positive five-point composite Newton–Cotes weights give

$$Q_w f = \sum_j \omega_{wj} f(q_j, v_j, u_j), \quad N_{wi,l} = Q_w \partial_i \chi_l, \quad A_{wi,ka} = Q_w[\psi_k S E_a S^{\mathsf{T}} v]_i.$$

Thus $C = [N, A] \in \mathbb{R}^{560\times 85}$, with columns $N_{40}$ followed by $A_{45}$, and with each $\psi_k$ followed by $E_1, E_2, E_3$. Windows are ordered by trajectory and increasing start index, then component $i = 1, 2$. All rows below use the original weak increment units. Any training row normalization must first be undone, with its rounding accounted for.

For $\Pi = Gv$, $g(q) = d\cos(d^{\mathsf{T}} q)$, define

$$\widehat{b}_{wi} = Q_w\{\tfrac{1}{2} v^{\mathsf{T}} G_i v - \partial_i \widehat{V} + u g_i\} - \Delta_w \Pi_i.$$

The implemented frozen-Hamiltonian base row is $-\widehat{b}$, not $\widehat{b}$. We denote by $h \in \mathbb{R}^{40}$ the finite nuisance vector written $b$ in the main text, and use $\Delta t$ for time increments. The unprojected finite residual of a matrix-valued learned head is

$$r_h = -\widehat{b} + Nh + A\widehat{c} + \widehat{t}. \tag{S54}$$

Here $\widehat{t}$ is the weak contribution of its nonpolynomial remainder. A floating-point projected loss is not substituted for $\|r_h\|$. The finite $h$ and this residual must be evaluated or enclosed independently.

**Lemma S1.** Reuse of the certified scaled inverse. *Let $C_0$ be the exact analytic matrix at the fixed nominal observations. Let $D$ be the fixed positive diagonal matrix of stored dyadic column scales and $L \in \mathbb{R}^{85\times 560}$ the fixed stored dyadic approximate inverse of $C_0 D^{-1}$. Suppose the interval certificate proves*

$$T_0 = L C_0 D^{-1}, \qquad \|T_0 - I\|_\infty \le \epsilon_0 < 1.$$

*For a new complete bundle with matrix $C_1$, suppose*

$$\left\|L(C_1 - C_0)D^{-1}\right\|_\infty \le \eta, \qquad \epsilon_0 + \eta < 1.$$

*Then $C_1$ has full column rank and the exact left inverse*

$$K_1 = D^{-1}(LC_1D^{-1})^{-1}L, \qquad K_1C_1 = I_{85}, \tag{S55}$$

*satisfies*

$$\|K_1\|_\infty \le \frac{\left\|D^{-1}\right\|_\infty \|L\|_\infty}{1-\epsilon_0-\eta}.$$

*For any row output map $O$, its gain is bounded more usefully by*

$$\|OK_1\|_\infty \le \frac{\left\|OD^{-1}\right\|_\infty \|L\|_\infty}{1-\epsilon_0-\eta}.$$

*In particular, with $\kappa_\infty = \left\|D^{-1}\right\|_\infty \|L\|_\infty / (1-\epsilon_0-\eta)$, a conservative Euclidean Gram bound is $C_1^{\mathsf{T}}C_1 \succeq (85\kappa_\infty^2)^{-1}I$. The transpose $K_1^{\mathsf{T}}$ is a right inverse of $C_1^{\mathsf{T}}$; $C_1$ itself has a left inverse.*

**Proof.** Writing $T_1 = LC_1D^{-1}$ gives $\|T_1 - I\|_\infty \le \epsilon_0 + \eta < 1$. The Neumann series yields $\left\|T_1^{-1}\right\|_\infty \le (1-\epsilon_0-\eta)^{-1}$. Multiplication proves (S55) and both gain estimates. Full column rank follows from $K_1C_1 = I$. Moreover $\|z\|_2 \le \sqrt{85}\,\|z\|_\infty \le \sqrt{85}\kappa_\infty \|C_1 z\|_2$, which proves the Gram bound.

This is directly compatible with the existing nominal certificate and its saved $D, L$. A newly whitened Gram matrix is unnecessary for this gate. Neither numerical singular values nor the rounded matrix centre replace the interval statement about the true entries of $C_0$.

*A computable complete-bundle perturbation budget.* Let $x = (q,p)$ and let $\varphi_i(q,v)$ be row $i$ of the $2 \times 85$ instantaneous feature matrix above. Put $\widetilde{\varphi}_i = \varphi_i D^{-1}$ and use the vector 1-norm for these rows. Fix a common canonical real tube $\Omega$ and a convex observation box containing its $(q, v(q,p))$ image and all compared observed nodes. On this observation box let

$$\|\widetilde{\varphi}_i(q,v) - \widetilde{\varphi}_i(q',v')\|_1 \le L_{q,i}\|q-q'\|_2 + L_{v,i}\|v-v'\|_2, \quad M_i \ge \sup \|\widetilde{\varphi}_i\|_1 .$$

The canonical observation map is $v(q,p) = G(q)^{-1}p$. For the stated reference, use the fixed metric

$$W = \begin{pmatrix} K_0 & (7/400)I_2 \\ (7/400)I_2 & G(0)^{-1} \end{pmatrix}, \qquad K_0 = B + dd^{\mathsf{T}}, \qquad \|x\|_W = (x^{\mathsf{T}}Wx)^{1/2}.$$

Its positivity is verified by an exact rational LDL factorization. The cross term specifies a norm; the physical damping remains $R_*(q)$. Set

$$\kappa_q = \left\|\mathrm{proj}_q W^{-1/2}\right\|_2, \quad L_v^W \ge \sup_\Omega \left\|D_x v\, W^{-1/2}\right\|_2, \quad L_{x,i} = L_{q,i}\kappa_q + L_{v,i}L_v^W .$$

All factors require outward bounds. In particular, an error bound for $q$ alone cannot bound the $A$ rows. The useful exact identity

$$SE_aS^{\mathsf{T}}v(q,p) = SE_aS^{-1}p$$

allows canonical feature derivatives to be bounded using $S$, $S^{-1}$, their first $q$ derivatives and the momentum range. For example,

$$\begin{aligned}\partial_j(\psi SE_aS^{-1}p) &= (\partial_j\psi)SE_aS^{-1}p \\ &\quad + \psi\{S_jE_aS^{-1} - SE_aS^{-1}S_jS^{-1}\}p, \\ \partial_p(\psi SE_aS^{-1}p) &= \psi SE_aS^{-1}.\end{aligned}$$

This identity does not identify a stored numerical $v_j$ with $G(q_j)^{-1}p_j$; that discrepancy is a separate measured/enclosed term.

For nominal trajectory $k$, let a true-reference IVP certificate at every stored time provide a canonical radius $R_{kj}$ in $W$ norm around its native centre $x_{kj}^n$. Define the observable-centre discrepancies

$$a_{kj}^q \ge \left\|q_{kj}^{\mathrm{obs}} - q_{kj}^n\right\|_2, \qquad a_{kj}^v \ge \left\|v_{kj}^{\mathrm{obs}} - v(x_{kj}^n)\right\|_2 .$$

Then a feature-error allowance between the stored observations and the nominal true flow is

$$e^0_{ki,j} = L_{q,i}a^q_{kj} + L_{v,i}a^v_{kj} + L_{x,i}R_{kj}.$$

Equivalently, directly certified $q$ and velocity observation errors can be used in the first two terms, without counting $R_{kj}$ twice.

Let each prospective initial condition satisfy $\|x^{\mathrm{new}}_k(0) - x_k(0)\|_W \le \delta_k$, with the same prescribed constant input. Assume the reference field's logarithmic norm is bounded by a fixed $\ell \ge 0$ throughout the connecting tubes. As long as these tubes stay in $\Omega$, Gronwall's inequality gives

$$\|x^{\mathrm{new}}_k(t) - x_k(t)\|_W \le e^{\ell t}\delta_k.$$

This domain premise needs an independent distance-to-boundary budget; using a derivative bound outside its certified tube would be circular. If new observations incur an additional feature-error allowance $e^1_{ki,j}$ and their stored time differs by at most $\vartheta_{kj}$, let $F_W \ge \sup_\Omega \|f_*\|_W$. With exact old and new quadrature weights, one sufficient row bound is

$$\begin{aligned} E_{kwi} = &\sum_j |\omega^1_{wj}|\{e^0_{ki,j} + L_{x,i}e^{\ell t_{kj}}\delta_k + L_{x,i}F_W\vartheta_{kj} + e^1_{ki,j}\} \\ &+ \sum_j |\omega^1_{wj} - \omega^0_{wj}|M_i, \\ \left\|L(C_1 - C_0)D^{-1}\right\|_\infty \le &\max_{a=1,\dots,85}\sum_{k,w,i}|L_{a,kwi}|E_{kwi} =: \eta. \end{aligned} \tag{S56}$$

In the time term the Gronwall factor must use the largest compared time if the time shift is positive. Setting that time to the common end $T$ is a simpler safe choice. The last inequality follows from the row 1-norm bound for $(C_1 - C_0)D^{-1}$ and entrywise absolute multiplication by $L$.

All mathematical times here are exact rationals represented by the observation array $\tau$. A raw SI clock multiplied by a separately rounded $\omega_0$ is not silently declared to be the same clock. If the prospective design enforces identical stored $\tau$ and weights, then $\vartheta = 0$ and the weight term vanishes; otherwise both terms must be certified. Finite-time quadrature remains part of the model, not an exact continuous integral.

For example, if common perturbations give the proved bound $\eta \le \eta_{\mathrm{base}} + A_\delta\delta$ after all timing, observation and arithmetic allowances are included, a positive radius can be chosen only after proving $\eta_{\mathrm{base}} < 1 - \epsilon_0$. It must satisfy both $A_\delta\delta < 1 - \epsilon_0 - \eta_{\mathrm{base}}$ and the connecting-tube distance condition. There is no positive-radius conclusion from rank alone.

**Corollary S2.** A certified local fourteen-trajectory design. *Retain the fourteen prescribed nominal initial conditions, the same finite-Gauss physical-viscosity reference with autonomous input $u = 0$, and the corresponding stored time labels and exact Newton–Cotes weights. For each trajectory, let the true initial perturbation satisfy*

$$\|\delta x_k(0)\|_W \le R_W = 2^{-34} = \frac{1}{17179869184}, \qquad k = 1, \dots, 14.$$

*At every paired node, suppose the new observations satisfy*

$$\max\{|q^{\mathrm{obs}}_{kji} - q^{\mathrm{true}}_{kji}|, |v^{\mathrm{obs}}_{kji} - v^{\mathrm{true}}_{kji}|\} \le \sigma = 10^{-11}, \qquad i = 1, 2.$$

*Then the resulting complete observation matrix $C_1$ satisfies*

$$\left\|L(C_1 - C_0)D^{-1}\right\|_\infty < 0.039651094 < \frac{1}{8}, \qquad 1 - \left\|LC_1D^{-1} - I\right\|_\infty > 0.960348906.$$

*Consequently the inverse in Lemma S1 is valid for every such complete bundle. The recovery estimates below additionally require their stated residual, nuisance and remainder hypotheses.*

**Proof.** The nominal centre trajectories have been enclosed at all $14 \cdot 801 = 11214$ nodes and throughout all $14 \cdot 800 = 11200$ slabs. The original first trajectory certificate is retained, and thirteen additional independent IVP certificates complete this collection. Their exact stored-label horizons are at most $T = 3602879701896397/9007199254740992$, the exact

binary64 value of $0.4$. The common $W$ and its exact inverse give the component bound $|\delta x_j| \le \sqrt{(W^{-1})_{jj}}\,\|\delta x\|_W$. The logarithmic-norm estimate therefore transports $R_W$ with $e^{\ell T}\sqrt{(W^{-1})_{jj}}$. For velocity, the component gains additionally retain

$$|\delta v_i| \le \sum_j U_{ij}|\delta p_j| + \sum_{k,j}(U_q)_{kij}\,\frac{1}{10}|\delta q_k|,$$

where $U$ and $U_q$ are the full accepted bounds for $G^{-1}$ and its first configuration derivatives on the canonical cube. Substitution into the exact feature/row gains yields the following outward summaries. All comparisons are made using their exact rational bounds; the total is summed before the displayed decimal rounding.

| Contribution | Certified upper bound |
|---|---|
| Nominal observed-to-true-flow discrepancy, $\eta_{\mathrm{base}}$ | $0.000101409$ |
| New observation error $\sigma$, $\eta_{\mathrm{noise}}$ | $0.007582392$ |
| True initial-condition perturbations, $\eta_{\mathrm{IC}}$ | $0.031967294$ |
| Total $\eta$ | $0.039651094$ |
| Nominal scaled-inverse defect $\epsilon_0$ | $4.282\,10^{-11}$ |
| Combined scaled-inverse defect $\epsilon_0 + \eta$ | $0.039651094$ |

The selected $R_W$ is the largest power of two at most one half of the minimum matrix and domain feasible bounds. The complete nominal true tubes, plus the transported perturbations, lie strictly inside the canonical cube $|q_i|, |p_i| < 1/10$. The additional $\sigma$ margin keeps observed $q$ in its feature box. The inequalities $|v_i| + \sigma \le (1/10)\sum_j U_{ij} + \sigma$ keep observed velocity strictly inside $|v_1| < 1/5$, $|v_2| < 4/5$. These domain checks also justify the perturbation estimate: apply it up to a possible first exit; the strict interior bound contradicts that exit. Equation (S56) and Lemma S1 give the claimed margin.

The radius and observation tolerance are sufficient conditions for high-accuracy numerical data in this small neighborhood and the stated nondimensional chart. The conclusion concerns observability of the specified $85$ finite output coordinates. It does not imply unique identification of neural-network weights realizing those outputs.

*Finite representatives, remainders and the nuisance coordinate.* For the new bundle assume, on $\mathcal{Q} = [-1/10, 1/10]^2$,

$$\widehat{V} - V_* = \chi^{\mathsf{T}} a_V + \rho_V, \qquad M_* = \sum c_{*,ka}\psi_k E_a + \rho_*, \qquad \widehat{M} = \sum \widehat{c}_{ka}\psi_k E_a + \widehat{\rho}.$$

Let $t_V = Q\nabla\rho_V$, $t_* = Q[S\rho_* S^{\mathsf{T}} v]$ and $\widehat{t} = Q[S\widehat{\rho}S^{\mathsf{T}} v]$. The signed reference consistency vector is

$$e_{wi} = Q_w\{\tfrac{1}{2}v^{\mathsf{T}} G_i v - \partial_i V_* + u g_i - [SM_* S^{\mathsf{T}} v]_i\} - \Delta_w (Gv)_i.$$

Exact subtraction, including (S54), gives

$$r_h = C_1 \begin{bmatrix} a_V + h \\ \widehat{c} - c_* \end{bmatrix} + \widehat{t} - t_* + t_V - e. \tag{S57}$$

Equivalently, the linear-response target for $C_1$ against $\widehat{b}$ is $[-a_V; c_*]$, while the fitted vector is $[h; \widehat{c}]$. These equivalent signs prevent profiling from being misread as training the conservative Hamiltonian.

Let $K_H = [I_{40}, 0]K_1$ and $K_M = [0, I_{45}]K_1$. By Lemma S1, $K_H N = I$, $K_H A = 0$, $K_M N = 0$ and $K_M A = I$. Suppose all potential representatives extend holomorphically to a common complex rectangle $\mathcal{R}$ and $\|\partial^\alpha \rho_V\|_{\mathcal{R}} \le R_\alpha$. Define output gains

$$\beta_{H,\alpha} \ge \sup_{\mathcal{R}} \left\|\partial^\alpha \chi^{\mathsf{T}} K_H\right\|_1, \qquad \beta_M \ge \sup_{\mathcal{Q}} \|F(q)K_M\|_{\infty\to 2},$$

where $F(q)c = (M_{11}, \sqrt{2}M_{12}, M_{22})$ for the polynomial matrix. Lemma S1 supplies conservative computable gain bounds; output-specific validated bounds may improve them. For $E \ge \|e\|_\infty$, $T_V \ge \|t_V\|_\infty$, $K_{\mathrm{weak}} \ge$

$\max_{w,i}\sum_j|\omega_{wj}|\sqrt{G_{ii}}\sqrt{v^{\mathsf{T}}Gv}$ and $\sup_{\mathcal{Q}}\|\rho_*\|_2\le\zeta_*$, $\sup_{\mathcal{Q}}\|\widehat{\rho}\|_2\le\widehat{\zeta}$, it follows that

$$\sup_{\mathcal{Q}}\left\|\widehat{M}-M_*\right\|_2\le\beta_M\{\|r_h\|_\infty+E+T_V+K_{\mathrm{weak}}(\zeta_*+\widehat{\zeta})\}+\zeta_*+\widehat{\zeta}, \tag{S58}$$

$$\begin{aligned}\left\|\partial^\alpha(\widehat{V}-V_*)\right\|_{\mathcal{R}}&\le\sup_{\mathcal{R}}\left|\partial^\alpha\chi^{\mathsf{T}}[K_H(r_h-\widehat{t})-h]\right|\\&\quad+\beta_{H,\alpha}(E+T_V+K_{\mathrm{weak}}\zeta_*)+R_\alpha=:B_\alpha.\end{aligned} \tag{S59}$$

For the second bound, $\widehat{t}$ can instead be bounded by $K_{\mathrm{weak}}\widehat{\zeta}$ when it is not evaluated with its sign. These inequalities follow by applying $K_H, K_M$ to (S57), then using $\|\mathrm{row}_i\,S\|_2=\sqrt{G_{ii}}$ and $\left\|S^{\mathsf{T}}v\right\|_2=\sqrt{v^{\mathsf{T}}Gv}$.

In particular $K_H$ recovers $a_V+h$; the deployed conservative error uses $a_V=(a_V+h)-h$. Small $r_h$ alone gives no small $a_V$ if $h$ is uncontrolled. In the damping-only fitting stage, the previously trained conservative Hamiltonian is fixed. Equation (S59) therefore certifies that fixed function; it is not an additional fit of $H$. A future fit with $h=0$ could use the simpler residual-only bound if it actually attains a certified residual and tail budget. Replacing deployment by $\widehat{V}+\chi^{\mathsf{T}}h$ is a different Hamiltonian and needs its own checkpoint and KAM validation. Only derivative orders one through three are needed for the KAM transport; an absolute potential-value estimate additionally requires a fixed energy gauge or a separately bounded constant remainder.

*Consistency and the conservative KAM threshold.* The discrete identity for $e$ is valid at arbitrary observed nodes. To bound $e$ through true trajectories, one must additionally include quadrature error for $\dot{p}$, the force-feature observation errors and the two endpoint errors of $G(q)v$. Alternatively, a direct signed enclosure of the displayed $e$ supplies $E$ without a continuous-quadrature claim. In either route, polynomial tails, raw-network evaluation error and stored finite $h$ remain in the budget.

For ordered derivatives on $\mathcal{R}$, define

$$\begin{aligned}\varepsilon_0&=\max(B_{10},B_{01}),\\\varepsilon_1&=\max(B_{20}+B_{11},B_{11}+B_{02}),\\\varepsilon_{t1}&=B_{20}+2B_{11}+B_{02},\\\varepsilon_2&=\max(B_{30}+2B_{21}+B_{12},B_{21}+2B_{12}+B_{03}).\end{aligned}$$

These respectively bound the conservative field difference, its Jacobian, the total Hessian entry sum and its ordered second derivative. The equality of the two kinetic metrics is essential here. They apply only to the unforced, nondissipative Hamiltonian pair.

A sufficient prospective learning-to-KAM gate is therefore:

1. (S56) proves $\epsilon_0+\eta<1$ for the complete bundle;
2. the frozen learned output, finite residual, nuisance and analytic tails prove (S59) through order three;
3. these ordered $\varepsilon$ bounds, together with the same-domain geometry, all-integer Diophantine and nondegeneracy bounds, satisfy every inequality of an independently validated conservative KAM theorem.

The last threshold is the full KAM gate, not a universal scalar tolerance on a real training loss. The existing native-observation transport already provides such a gate for its exact frozen pair, frequency and response map. For a different learned Hamiltonian, its cached Fourier residuals or cubic response coefficients cannot be reused merely because the new loss is small; the new residual/domain/nondegeneracy premises must be verified or transported with explicit error bounds. Analytic tails on $\mathcal{R}$ are also necessary: real-node recovery does not itself control a complex strip.

*Scope of the finite-output gate.* The fourteen nominal centre-trajectory IVP and velocity enclosures and the positive-radius sufficient-design gate are now available, as used in Corollary S2. No new perturbed bundle has been generated or fitted to obtain this conclusion. An accepted new bundle must still satisfy the prescribed support, fixed clock, exact quadrature and observation-error conditions. Its learned output must then meet the residual, finite-nuisance and analytic-tail bounds in (S58)–(S59), followed by the complete conservative KAM gates for the appropriate frozen function. The established native-source KAM result for the original fixed pair is unchanged. The bundle condition does not guarantee the success of an arbitrary training procedure, instantiate an iid statistical learning theorem, or establish a general advantage of complex networks.

## B Exact KAM inputs and certificate provenance

The certified frequency is the period-one frequency $\nu = \omega/(2\pi)$, where

$$\omega_1 = \frac{499974648628101}{500000000000000}, \qquad \omega_2 = \omega_1[6; 6, 1, 1, 1, 2, 1, 5, 14, 17, 4, \overline{1}].$$

The continued fraction has convergents $(p_n, q_n) = (2524851, 410497)$ and preceding pair $(622102, 101143)$, followed by the infinite golden tail. Its bounded partial quotients provide the all-integer small-divisor certificate used with $\gamma_D = 4828522443384859/576460752303423488$ and exponent one. The exact rational inputs below are drawn from the native-observation KAM-transport certificate; displayed residuals in the article are rounded outward from that certificate.

**Table S2.** Selected exact scalar inputs for the transported canonical a posteriori theorem. The norm conventions and all eight tests are retained in the machine-readable certificate.

| Quantity | Exact input |
|---|---|
| Input and final strip | $1/40, \quad 1/400$ |
| Domain margin | $3/1000$ |
| $DK$ bound | $36774512869/125000000000$ |
| $(DK)^{\mathsf{T}}$ bound | $73540395727/125000000000$ |
| Gram-inverse bound | $7951814956883/500000000000$ |
| Mean-torsion-inverse bound | $9890129091/500000$ |
| Field bound | $192221445399/500000000000$ |
| Jacobian bound | $6365861788341/1000000000000$ |
| Total first-derivative bound | $3975706639141/200000000000$ |
| Second-field-derivative bound | $1133437451689/500000000000$ |

The complete recomposed residual is less than $1.205897 \times 10^{-41}$, the sufficient threshold exceeds $5.82604 \times 10^{-38}$, and the strict smallness quotient is less than $0.000206984$. These scalars apply to the same frozen checkpoint, Hamiltonian homotopy, frequency and cubic response. They do not certify a generic error ball or a different fitted network.

All machine-readable results used in the article are linked to their source files by SHA-256 in the local verification manifest. The data map distinguishes the two training cohorts, the single reserved-data evaluation for the selected matrix comparison, native solver exports, independent reference-IVP validation and interval/rational certificates. The proof uses outward enclosure semantics and exact stored binary64 constants. Independent audits recheck source bindings, scalar inequalities, domain/partition coverage and selected exact arithmetic. They do not constitute a second implementation of every interval elementary operation or the complete ODE verifier. The enclosure library, arithmetic semantics and admitted computational conditions remain explicit premises of the numerical proof.

## C Complex-gradient descent with a strict descent margin

Theorem 3 in the published version of Zhang et al. (2022) motivates complex-gradient optimization, but a stationary-point conclusion requires a strict descent margin. The following convention-explicit statement also applies to a smooth CHNN regression objective.

**Proposition S3.** Complex descent. *Let $\mathcal{L} : \mathbb{C}^P \simeq \mathbb{R}^{2P} \to \mathbb{R}$ be bounded below and have an $L$-Lipschitz real gradient on every segment traversed by the iterates. For $w = a + ib$, define $G(w) = \nabla_a \mathcal{L} + i\nabla_b \mathcal{L} = 2\partial\mathcal{L}/\partial\bar{w}$. Choose a constant $\rho \in \mathbb{C}$ with $\kappa = \operatorname{Re}\rho - (L/2)|\rho|^2 > 0$, and update $w_{t+1} = w_t - \rho G(w_t)$. Then*

$$\mathcal{L}(w_{t+1}) \le \mathcal{L}(w_t) - \kappa\|G(w_t)\|_2^2, \qquad \lim_{t\to\infty} \|G(w_t)\|_2 = 0,$$

*and*

$$\min_{0\le t<T} \|G(w_t)\|_2^2 \le \frac{\mathcal{L}(w_0) - \inf \mathcal{L}}{\kappa T}.$$

**Proof.** Under realification, multiplication by $\rho = u + iv$ is $uI + vJ_c$, with $J_c^\top = -J_c$ and $J_c^\top J_c = I$. Hence the update has inner product $-u\|G\|^2$ with the real gradient and squared norm $|\rho|^2\|G\|^2$. The real descent lemma gives the first inequality. Summing it and using the lower bound for $\mathcal{L}$ gives $\kappa \sum_{t\ge 0} \|G(w_t)\|^2 \le \mathcal{L}(w_0) - \inf \mathcal{L}$. Square summability

proves convergence of the gradient norm and the minimum bound. The result asserts convergence to stationarity in gradient norm, not convergence to a global minimum or to a particular parameter vector.

For a real step this condition is $0 < \rho < 2/L$ under the stated convention. The case $\rho = 0$, allowed by a non-strict printed step-size interval, cannot imply gradient convergence. Nonsmooth complex activations require separate treatment. The present literal complex $\tanh$ networks are smooth as real maps wherever their poles are excluded. On a bounded parameter region with a uniform pole margin, the smooth regression objective admits finite local Lipschitz bounds. Applying the proposition to a training run would still require a verified common iterate region and the strict step-size margin above.

This realification argument is an optimization component motivated by Zhang et al. (2022); it does not transfer that paper's exponential-loss learning conclusions to the present mechanical regression. The recorded conservative fit uses full-batch Armijo steps on the real and imaginary parameter coordinates. The subsequent dissipation-head fit uses damped Gauss–Newton steps with Armijo search and retains its finite potential nuisance coordinate. Neither recorded algorithm is the constant-complex-step iteration of this proposition. Their finite loss-decrease records therefore do not establish the proposition's gradient-norm convergence conclusion. The identification-error and a-posteriori KAM results are independent of such convergence: they do not assume that the accepted frozen parameters are stationary or globally optimal.

## D A conditional learning-to-analytic-KAM implication

We relate the energy generalization estimate of Chen et al. (2022) to the analytic invariance residual required by Haro and Luque (2019, Theorem 3.1). The argument combines an interior gradient estimate with interpolation on a complex torus strip. This conditional statistical route requires hypotheses on the sampling distribution and the entire hypothesis class. A deterministic a-posteriori certificate for an individual learned Hamiltonian has a distinct scope: it establishes the existence of an invariant torus of that Hamiltonian without a population-risk or generalization statement.

### *D.1 The conditional statistical input and its energy gauge*

Work in one fixed canonical coordinate and time normalization, with phase dimension $m = 2n$ and $n \geq 2$. Let $D \subset \mathbb{R}^m$ be a bounded, connected Lipschitz domain. The conservative target is $H_\star$, and $\mathcal{H} = \{H_\vartheta : \vartheta \in \Theta\}$ is a hypothesis class fixed before the data are drawn. All energies are real valued on $D$, even when the network parameters are complex. Fix a finite exponent $p > m$ and a confidence parameter $0 < \alpha < 1$. Assume $H_\star, H_\vartheta \in W^{1,p}(D)$, using their continuous representatives, and define the pointwise conservative loss

$$\ell_\vartheta(x) = \|\nabla H_\vartheta(x) - \nabla H_\star(x)\|_p^p. \tag{S60}$$

Assume that $X_1, \ldots, X_N$ are independent and identically distributed with law $P$ supported on $D$, whose density relative to Lebesgue measure satisfies $f_P \geq m_P > 0$. Assume $0 \leq \ell_\vartheta \leq L_{\max}$ uniformly on the entire class and domain. Let $\overline{\mathfrak{R}}_N$ be a valid upper bound for the empirical Rademacher complexity of this loss class, with the normalization in Lemma 2 of Chen et al. (2022). Define

$$\begin{aligned} \widehat{L}_{p,N}(\vartheta) &= \frac{1}{N}\sum_{i=1}^{N} \ell_\vartheta(X_i), \\ A_N(\vartheta, \alpha) &= \widehat{L}_{p,N}(\vartheta) + 2\overline{\mathfrak{R}}_N + 3L_{\max}\sqrt{\frac{2\log(4/\alpha)}{N}}. \end{aligned} \tag{S61}$$

Under these assumptions the cited generalization bound supplies an event $\mathcal{G}_\alpha$, with $\mathbb{P}(\mathcal{G}_\alpha) \geq 1 - \alpha$, on which $\mathbb{E}_P \ell_\vartheta \leq A_N(\vartheta, \alpha)$ simultaneously for every $\vartheta \in \Theta$. This uniformity is essential when the same data select the fitted parameters.

Define the gauge-fixed error

$$e_\vartheta = H_\vartheta - H_\star - c_\vartheta, \qquad c_\vartheta = \frac{1}{|D|}\int_D (H_\vartheta - H_\star)\, dx. \tag{S62}$$

Let $C_P$ satisfy $\|e\|_{L^p(D)} \le C_P \|\nabla e\|_{L^p(D)}$ for mean-zero functions, and let $C_S$ be a valid constant in $W^{1,p}(D) \hookrightarrow C^0(\overline{D})$, using $\|e\|_{W^{1,p}}^p = \|e\|_{L^p}^p + \|\nabla e\|_{L^p}^p$.

**Proposition S4.** Conditional energy bound. *On $\mathcal{G}_\alpha$, simultaneously for every member of the class,*

$$\begin{aligned} \|e_\vartheta\|_{C^0(\overline{D})} &\le \varepsilon_\vartheta, \\ \varepsilon_\vartheta &:= C \left( \frac{A_N(\vartheta,\alpha)}{m_P} \right)^{1/p}, \\ C &= C_S (1 + C_P^p)^{1/p}. \end{aligned} \tag{S63}$$

**Proof.** The density lower bound and the uniform risk event give

$$\int_D \|\nabla e_\vartheta\|_p^p \, dx \le m_P^{-1} \mathbb{E}_P \ell_\vartheta \le m_P^{-1} A_N(\vartheta,\alpha).$$

Poincare's inequality applies because (S62) has zero mean. Consequently $\|e_\vartheta\|_{W^{1,p}} \le (1 + C_P^p)^{1/p} (A_N/m_P)^{1/p}$. The Sobolev embedding, valid because $p > m$, proves the assertion.

The exponent $1/p$ in (S63) follows from the $p$th-power definition of the loss (S60); a linear dependence on that loss would require different definitions. For the four-dimensional phase space considered here, this first-order embedding requires $p > 4$. Here $p$ is the integrability exponent in $W^{1,p}$, not a derivative order. Using weak-increment observations or a combined Hamiltonian–damping residual to bound $\widehat{L}_{p,N}$ additionally requires conservative-gradient label-error estimates, including errors in dissipation and input subtraction. The trajectory sampling scheme, with consecutive and overlapping windows, does not meet the iid state-sampling hypothesis of Proposition S4.

## *D.2 A real gradient bound with explicit domain loss*

**Proposition S5.** Interior gradient estimate from energy values. *Let $e \in C^2(D)$ satisfy $\sup_D |e| \le \varepsilon$ and $\max_j \sup_D |\partial_{jj} e| \le B$, with $B \ge 0$. Let $S \subset D$ and $r > 0$ satisfy $x + [-r, r]\mathbf{e}_j \subset D$ for every $x \in S$ and coordinate unit vector $\mathbf{e}_j$. Then*

$$\sup_{x \in S} \|\nabla e(x)\|_\infty \le b_r(\varepsilon, B) := \inf_{0 < h \le r} \left( \frac{\varepsilon}{h} + \frac{Bh}{2} \right). \tag{S64}$$

*For $B > 0$ this is*

$$b_r(\varepsilon, B) = \begin{cases} \sqrt{2B\varepsilon}, & 0 \le \varepsilon \le Br^2/2, \\ \varepsilon/r + Br/2, & \varepsilon > Br^2/2. \end{cases} \tag{S65}$$

*For $B = 0$, take $b_r(\varepsilon, 0) = \varepsilon/r$.*

**Proof.** For $0 < h \le r$, the fundamental theorem of calculus gives

$$\begin{aligned} &\left| \partial_j e(x) - \frac{e(x + h\mathbf{e}_j) - e(x - h\mathbf{e}_j)}{2h} \right| \\ &\quad \le \frac{1}{2h} \int_{-h}^{h} |\partial_j e(x + t\mathbf{e}_j) - \partial_j e(x)| \, dt \le \frac{Bh}{2}. \end{aligned}$$

The difference quotient has magnitude at most $\varepsilon/h$. Taking the infimum proves (S64). For $B\varepsilon > 0$, the unconstrained minimizer is $h = \sqrt{2\varepsilon/B}$; otherwise use $h = r$ or take the appropriate zero limit. This proves (S65).

The estimate only needs bounds on the pure second derivatives of the *error*; a full Hessian operator bound is a stronger sufficient condition. A certified sum of target and network Hessian bounds also suffices. An additive energy constant has no effect on this step.

Let $J = \left( \begin{smallmatrix} 0 & I_n \\ -I_n & 0 \end{smallmatrix} \right)$, and fix a reference embedding $K_0 : \mathbb{T}^n \to D$ and frequency $\omega \in \mathbb{R}^n$. Suppose its real image has the coordinate-segment margin in Proposition S5, and that an independently justified target residual obeys

$$\|J \nabla H_\star(K_0) - DK_0 \omega\|_0 \le \delta_\star. \tag{S66}$$

The subscript 0 denotes the supremum over real angles, with the maximum norm for vector components. Since $\|J\|_{\infty\to\infty} = 1$, the learned invariance residual satisfies

$$E_\vartheta = J\nabla H_\vartheta(K_0) - DK_0\omega,$$
$$\|E_\vartheta\|_0 \le \delta_\star + b_r(\varepsilon_\vartheta, B) =: \delta_\vartheta \quad \text{on } \mathcal{G}_\alpha. \tag{S67}$$

In particular, a certificate for an invariant torus of $H_\vartheta$ cannot be substituted for the target residual in (S66). If $K_0$ is an exact torus of $H_\star$, then $\delta_\star = 0$; otherwise its defect must be included.

The weak-increment objective also admits a deterministic application. Suppose the deterministic observation theorem leading to (S13) supplies first-derivative bounds for the same frozen $H_\vartheta, H_\star$ on the real image $K_0(\mathbb{T}^n)$. Taking their component maximum, as in (S18), gives $G_{\mathrm{det}}$ and hence $\|E_\vartheta\|_0 \le \delta_\star + G_{\mathrm{det}}$ directly. The identical known kinetic metric in the two Hamiltonians is retained in this use of the deterministic theorem. Substitution of $G_{\mathrm{det}}$ for $b_r(\varepsilon_\vartheta, B)$ in the analytic step below gives a deterministic residual bound without a statistical assumption. Its applicability requires the stated finite-nuisance, reference-consistency, class-remainder and common-domain estimates; a small floating-point profiled loss does not supply these hypotheses. No iid probability interpretation is attached to correlated windows.

### *D.3 Analytic continuation of the residual with a certified outer bound*

Write $\mathbb{T}^n_R = \{\theta \in \mathbb{C}^n/\mathbb{Z}^n : |\operatorname{Im}\theta|_\infty < R\}$. For a vector function, $\|F\|_R = \max_j \sup_{\mathbb{T}^n_R} |F_j|$. Assume that $K_0$ extends holomorphically to this strip and that $K_0(\mathbb{T}^n_R)$ lies in a pole-free complex domain of $H_\vartheta$. Assume a *certified upper bound*

$$\|E_\vartheta\|_R \le M. \tag{S68}$$

For example, if verified constants satisfy $\sup_{K_0(\mathbb{T}^n_R)} \|J\nabla H_\vartheta\|_\infty \le C_X$ and $\|DK_0\|_R \le C_K$, then one may *choose* $M = C_X + C_K\|\omega\|_\infty$. Here the matrix norm is $\|A\|_R = \max_i \sum_j \sup_{\mathbb{T}^n_R} |A_{ij}|$, as in the analytic norm convention of Haro and Luque (2019). Point samples of a complex extension do not certify these suprema.

**Lemma S2.** Residual interpolation on a torus strip. *Let $F$ be a bounded holomorphic periodic vector function on $\mathbb{T}^n_R$ with $\|F\|_0 \le \delta$ and $\|F\|_R \le M$. For $0 < \rho < R$, set $\bar\delta = \min\{\delta, M\}$. Then*

$$\|F\|_\rho \le \bar\delta^{\,1-\rho/R} M^{\rho/R}. \tag{S69}$$

*If either bound is zero, the right-hand side is interpreted by continuity.*

**Proof.** First suppose $M \ge \bar\delta > 0$ and that the extension is continuous on the closed strip. Given $x + iy$ with $t = |y|_\infty > 0$, put $v = y/t$. For each component define the one-variable function $g(w) = F_j(x + wv)$ on $0 \le \operatorname{Im} w \le R$. The real parts $x + (\operatorname{Re} w)v$ range over real angles, so the lower boundary has bound $\bar\delta$ and the upper boundary has bound $M$. The function is bounded throughout this strip. Set

$$\beta = R^{-1}\log(M/\bar\delta), \qquad G(w) = \bar\delta^{-1} g(w) e^{i\beta w}.$$

Both horizontal boundaries satisfy $|G| \le 1$. To justify the maximum principle on this unbounded domain, use $G_\eta(w) = G(w)\exp(-\eta w^2 - \eta R^2)$, with $\eta > 0$. Its modulus is at most one on the horizontal boundaries and tends to zero on the vertical boundaries of expanding rectangles. The rectangle maximum principle, followed by expansion and then $\eta \downarrow 0$, gives $|G(w)| \le 1$. In particular,

$$|F_j(x + iy)| = |g(it)| \le \bar\delta \exp(\beta t) = \bar\delta^{1-t/R} M^{t/R}.$$

Since $\bar\delta \le M$, this expression is nondecreasing in $t$. Taking $t \le \rho$ and the componentwise supremum proves the claim. For an open outer strip, apply the argument with $\rho < R' < R$ and let $R' \uparrow R$. Zero bounds follow by replacing the bounds with positive ones and taking a limit. This also covers real angles $t = 0$.

Combining the energy and gradient estimates with the strip lemma gives, on $\mathcal{G}_\alpha$, the analytic residual bound

$$\begin{aligned} \|E_\vartheta\|_\rho &\le \mathcal{E}_\vartheta, \\ \mathcal{E}_\vartheta &:= \bar{\delta}_\vartheta^{1-\rho/R} M^{\rho/R}, \\ \bar{\delta}_\vartheta &:= \min\{M, \delta_\vartheta\}, \end{aligned} \tag{S70}$$

where $\delta_\vartheta$ is defined by (S67). There is no dimension factor for this maximum-component norm.

For the real energy readout of the complex network, analytic continuation means complexification of the real canonical variables, not application of a real-part operation at complex arguments. If $H(x) = \mathrm{Re}\, F(x)$ on real $x$ and $F$ has a holomorphic extension, its extension is $[F(z) + F^{\#}(z)]/2$, where $F^{\#}(z) = \overline{F(\bar{z})}$ is a holomorphic function obtained by conjugating coefficients. Features written as $(z, \bar{z})$ on real phase space become the independent linear expressions $(Q + iP)/\sqrt{2}$ and $(Q - iP)/\sqrt{2}$ after complexification. Intermediate complex $\tanh$ layers must remain separated from their poles on the whole declared complex domain. Bounds for one trained network do not establish corresponding uniform bounds for a parameter class.

### D.4 The precise a-posteriori KAM endpoint

We use the ordinary theorem of Haro and Luque (2019, Theorem 3.1, equations (3.9)–(3.11), arXiv v1), specialized to $d = n \ge 2$ and canonical geometry. This specialization has no additional first integrals; their constants are zero as in the cited Remark 3.2. In this paragraph $\mathcal{B}$ denotes the complex phase domain and is distinct from the scalar Hessian bound $B$. The following are the corresponding H1–H5 requirements, imposed independently of the empirical training loss:

- H1. The exact symplectic and metric structures and $H_\vartheta$ extend analytically to $\mathcal{B}$, with the specified analytic bounds. In canonical coordinates the geometric structures are constant, as in the cited Remark 3.3. The required Hamiltonian bounds concern $\|DH_\vartheta\|_{\mathcal{B}}$, $\|X_{H_\vartheta}\|_{\mathcal{B}}$, $\|DX_{H_\vartheta}\|_{\mathcal{B}}$, $\|D^2 X_{H_\vartheta}\|_{\mathcal{B}}$ and $\|D(X_{H_\vartheta}^{\mathsf{T}})\|_{\mathcal{B}}$ in the cited tensor-norm convention. As emphasized in its Section 3.1, this last derivative of a row-vector function and the transpose of the Jacobian have different tensor-norm definitions and are not interchangeable. The optional first integral $c$ of that theorem may be chosen constant, with zero derivative bound.
- H2. The embedding $K_0$ and $DK_0$ extend continuously to the closed strip of width $\rho$, are holomorphic inside it, and satisfy $\mathrm{dist}(K_0(\mathbb{T}^n_\rho), \partial\mathcal{B}) > 0$, $\|DK_0\|_\rho < \sigma_K$ and $\|(DK_0)^{\mathsf{T}}\|_\rho < \sigma_{K^{\mathsf{T}}}$.
- H3. The tangent frame $L = DK_0$ has maximal rank throughout the closed complex strip, and its metric Gram inverse $\mathsf{B} = (L^{\mathsf{T}}L)^{-1}$ satisfies $\|\mathsf{B}\|_\rho < \sigma_B$. The transpose is bilinear, not a Hermitian adjoint. The normal frame is defined by equations (3.2)–(3.5) of the cited theorem, with the same symplectic sign convention.
- H4. For the resulting torsion matrix $T$ of equations (3.6)–(3.7), computed using $X_{H_\vartheta}$, the real-angle average has an inverse satisfying $|\langle T\rangle^{-1}| < \sigma_T$. Small gradient error by itself does not establish this condition.
- H5. For fixed $\gamma > 0$ and $\tau \ge n - 1$, the frequency obeys $|k \cdot \omega| \ge \gamma |k|_1^{-\tau}$ for every $k \in \mathbb{Z}^n \setminus \{0\}$. A finite search for small divisors does not prove this all-integer assertion.

For $0 < \rho_\infty < \rho$, these data determine the explicit theorem constants $C_1, C_2, C_3$ of the cited Appendix B. Its smallness condition is exactly

$$\frac{C_1 \|E_\vartheta\|_\rho}{\gamma^4 \rho^{4\tau}} < 1. \tag{S71}$$

Accordingly, checking

$$\frac{C_1 \mathcal{E}_\vartheta}{\gamma^4 \rho^{4\tau}} < 1 \tag{S72}$$

and H1–H5 supplies, on $\mathcal{G}_\alpha$, an invariant torus $K_\infty$ for the learned conservative Hamiltonian with the prescribed frequency, analytic on the strip of width $\rho_\infty$. The cited theorem also gives

$$\|K_\infty - K_0\|_{\rho_\infty} \le \frac{C_2 \|E_\vartheta\|_\rho}{\gamma^2 \rho^{2\tau}} \le \frac{C_2 \mathcal{E}_\vartheta}{\gamma^2 \rho^{2\tau}}. \tag{S73}$$

The zero-residual case is already invariant and admits $K_\infty = K_0$; the weak inequality also covers this case. The constants and geometric bounds must correspond to this Hamiltonian, domain and pair of strip widths. The conclusion concerns a

conservative torus, not attraction, dissipative control, all trajectories, or a positive-measure family without an additional family argument.

**Proposition S6.** Deterministic certification of an arbitrary training output. *Let* $\mathsf{A}$ *be any possibly randomized training and checkpoint-selection procedure, and let* $\widehat{\vartheta} = \mathsf{A}(\mathcal{Z}, \xi)$ *be its finite output parameters, including data-dependent preprocessing constants. For each realized dataset* $\mathcal{Z}$ *and random seed* $\xi$*, consider the exact exported Hamiltonian* $H_{\widehat{\vartheta}}$ *and any candidate* $K_0$*, frequency* $\omega$ *and verification domain. Suppose these objects satisfy H1–H5 above in the same coordinate, time and norm conventions, and a valid deterministic bound* $\overline{E} \geq \|J\nabla H_{\widehat{\vartheta}}(K_0) - DK_0\omega\|_\rho$ *satisfies*

$$\frac{C_1\overline{E}}{\gamma^4\rho^{4\tau}} < 1.$$

*Then* $H_{\widehat{\vartheta}}$ *has the invariant torus and correction bound stated above, with* $\overline{E}$ *in place of* $\mathcal{E}_\vartheta$*. No sampling-independence, descent, stationarity or optimality condition on the training output is required.*

**Proof.** Fix an arbitrary realization $(\mathcal{Z}, \xi)$ and its exact output. The verified hypotheses and residual bound satisfy Haro and Luque (2019, Theorem 3.1), which gives the asserted torus and correction estimate. This implication holds for every realization; no property of the procedure producing the candidate is used.

This result concerns acceptance of a particular frozen function. It requires verified bounds for that function, rather than class-wide statistical constants, and does not guarantee that the training procedure produces an acceptable candidate. The finite-sample weak-loss bounds and conservative-error identities likewise apply to arbitrary selected parameters under their stated observation and model hypotheses.

### *D.5 An explicit risk threshold and the probability statement*

For $B > 0$, $M > 0$, set

$$s = \rho/R, \qquad \kappa = \frac{\gamma^4\rho^{4\tau}}{C_1}, \qquad T_0 = \left(\frac{\kappa}{M^s}\right)^{1/(1-s)}. \tag{S74}$$

If $T_0 > \delta_\star$, define the positive admissible energy error

$$\varepsilon_{\mathrm{allow}} = \min\left\{\frac{Br^2}{2}, \frac{(T_0 - \delta_\star)^2}{2B}\right\}. \tag{S75}$$

A sufficient empirical-risk and complexity condition is

$$A_N(\vartheta, \alpha) < m_P\left(\frac{\varepsilon_{\mathrm{allow}}}{C}\right)^p. \tag{S76}$$

Indeed, (S63) then yields $\varepsilon_\vartheta < \varepsilon_{\mathrm{allow}} \leq Br^2/2$ and hence $\delta_\star + \sqrt{2B\varepsilon_\vartheta} < T_0$. Raising to $1 - s$ and multiplying by $M^s$ proves (S72). The general criterion (S72) remains available when $B = 0$ or the optimal difference step is limited by $r$; these cases are covered directly by (S72).

**Corollary S3.** Uniform event and post-fit acceptance. *Suppose the statistical assumptions preceding Proposition S4 hold. Let* $\widehat{\vartheta}$ *be any data-dependent selection from the fixed class. Define acceptance to require valid deterministic bounds for all analytic, reference-residual and geometric assumptions above, together with* (S72) *evaluated using* (S63)*. Let* $\mathcal{A}$ *be this acceptance event and* $\mathcal{K}$ *the event that the asserted learned invariant torus exists. Then*

$$\mathbb{P}(\mathcal{A} \cap \mathcal{K}^c) \leq \alpha. \tag{S77}$$

**Proof.** On the uniform event $\mathcal{G}_\alpha$, the energy estimate applies to every parameter, including $\widehat{\vartheta}$. All subsequent implications are deterministic whenever the acceptance conditions hold. Thus erroneous acceptance is contained in $\mathcal{G}_\alpha^c$, whose probability is at most $\alpha$.

Equation (S77) is not the claim $\mathbb{P}(\mathcal{K} \mid \mathcal{A}) \geq 1 - \alpha$. If the acceptance probability is positive, the immediate conditional bound is instead at most $\alpha/\mathbb{P}(\mathcal{A})$ for failure. An unconditional success probability of at least $1 - \alpha$ requires the remaining

conditions to hold for the selected model; a procedure that can reject that model gives only the joint-event guarantee above. Application of this statistical result requires uniform parameter-class bounds, a valid complexity estimate for the realified complex architecture, iid state samples with a positive density lower bound, and conservative pointwise label-error estimates. The correlated weak-window sampling scheme does not satisfy the iid hypothesis. Moreover, bounds for the fitted Hamiltonian are model-specific and do not provide the class-wide complexity or distributional constants required in (S61). The numerical trajectory-based results are therefore interpreted through their deterministic estimates, without a population-confidence claim for the fitted model. The conditional statistical learning-to-KAM route in this section has not been instantiated by the present experiments.

**Remark S1.** Necessity of quantitative regularity bounds. Real energy closeness alone does not control a vector field: on a bounded real interval, $e_k(q) = k^{-1}\sin(k^2 q)$ has sup norm tending to zero and derivatives of unbounded amplitude. Even a real Hessian bound does not supply (S68): the entire, $2\pi$-periodic functions $e_k(q) = k^{-2}\sin(kq)$ have $\|e_k\|_0 \to 0$, $\|e_k'\|_0 \to 0$ and $\|e_k''\|_0 \le 1$, while $|e_k'(i\rho)| = k^{-1}\cosh(k\rho)$ diverges at any fixed $\rho > 0$. A pole-free extension together with a quantitative outer bound, not analyticity alone, is what makes Lemma S2 useful. These examples concern scalar functions and do not assert unboundedness within a separately constrained finite network class.

## E Uniform learning bounds and the raw parameter class

The generalization-to-KAM implication requires assumptions on an entire hypothesis class. In particular, Lemma 2 of Chen et al. (2022) assumes a bounded loss and iid samples from a fixed law $P$, with joint law $P^N$; Theorem 8 additionally assumes bounded layer norms and controlled activation derivatives. Its Assumption 1 supplies a positive lower bound for a sampling density, and the first-order Sobolev step uses an integrability exponent greater than four in the present phase dimension. These are distinct from a certificate for one frozen output.

Fix the prescribed metric, quartic prior, seven coefficient networks, normalizations and correction amplitude $\eta_V$. Write $P_\theta = 252$ for the number of raw real coordinates, and interpret the stored architecture as an exact function of $\theta \in \mathbb{R}^{P_\theta}$. The imaginary-row normalization is retained throughout this section. In particular, no activation or effective-weight formula is changed. Put

$$\ell_\theta(q,p) = \|\nabla H_\theta(q,p) - \nabla H_*(q,p)\|_6^6.$$

The kinetic terms agree, so this loss depends only on $q$. The unboundedness assertions that follow concern the continuous raw-parameter class. A class of finite binary64 words is finite after invalid outputs are excluded, but does not thereby acquire a useful envelope or a common prescribed complex domain.

**Proposition S7.** Real-strip guards do not bound the full raw class. *The current row constraints do not provide a finite class-uniform bound for $\ell_\theta$ on any configuration box with nonempty interior. For any finite sample containing a nonzero configuration, the empirical Rademacher complexity of this unbounded loss class is infinite.*

**Proof.** Choose the coefficient multiplying $q_2^6$. Set its first-layer raw weights and biases to zero; set the second-layer weights to zero and its first real bias to one. Give the first readout coefficient the real value $t$, and set all other readout coefficients to zero. The network output is the constant $t\tanh(1)$. The deployed coefficient uses $\mathrm{raw}$, rather than the zero-gauged residual, so even symmetrization retains this constant. All imaginary preactivations are zero, for every $t$. Thus this legal family has

$$V_t(q) = V_4(q) + \eta_V t\tanh(1) q_2^6, \qquad \partial_2(V_t - V_*) = 6\eta_V t\tanh(1) q_2^5 + \partial_2(V_4 - V_*).$$

At any $q_2 \neq 0$ its loss grows as a positive multiple of $t^6$. If a sample has only a nonzero first coordinate, use the $q_1^6$ coefficient instead. On the all-positive Rademacher-sign event, which has probability $2^{-N} > 0$, the supremum of the signed loss sum is infinite. For every other sign sequence it is bounded below by the finite value at any fixed parameter. Its expected supremum is therefore infinite.

For finite digital parameters the same example gives arbitrarily large scales up to the admitted word range, rather than an infinite cardinality or an infinite mathematical supremum. Even the member $t = 1$, with every raw coordinate at most one, already gives a useful quantitative obstruction below. Neither regularization in an objective nor a post-fit norm replaces a predeclared class-uniform learning premise.

**Proposition S8.** No common prescribed complex strip. *The same raw class contains potentials with genuine poles arbitrarily close to the real configuration domain, despite satisfying the real-state imaginary-row constraints.*

**Proof.** In the coefficient of $q_1^6$, retain one active path. Its first real weight is $t > 0$, its second raw real weight is one, its second real bias is one, and its readout is one. All other weights and biases are zero. Let $s_0$ be the stored input scale $\mathrm{binary64}(\sqrt{2})$, let $v_0$ be the fixed second-layer previous-imaginary bound, and set $\epsilon_0 = \sqrt{\mathrm{binary64}((10^{-12})^2)}$. The exact effective second weight is the positive constant

$$a_* = \frac{\rho_2}{\rho_2 + v_0\sqrt{1+\epsilon_0^2} + (3+v_0)\epsilon_0}.$$

The active branch and its even coefficient are

$$F_t(Q) = \tanh(1 + a_* \tanh(tQ_1/s_0))\,, \qquad c_t^{\mathrm{ev},\mathbb{C}}(Q) = \tfrac{1}{2}[F_t(Q) + F_t(-Q)].$$

All effective coefficients are real, so the sharp extension equals $F_t$ and introduces no further cancellation. Let

$$z_* = \frac{-1 + i\pi/2}{a_*}, \qquad Q_t = \Big(\frac{s_0}{t}\operatorname{arctanh} z_*,\, 0\Big)\,.$$

The principal inverse is finite here and satisfies $\tanh(\operatorname{arctanh} z_*) = z_*$. The second preactivation of $F_t(Q_t)$ is $i\pi/2$, a pole of $\tanh$, whereas that of $F_t(-Q_t)$ is $2 - i\pi/2$, where $\cosh$ is nonzero. The inner derivative is nonzero since $z_* \neq \pm 1$. Consequently the even coefficient has a nonremovable simple pole. Also $Q_{t,1} \neq 0$, so multiplication by $Q_1^6$ cannot remove it. Finally $Q_t \to 0$ as $t \to \infty$.

For the exact current constants, outward evaluation with $t = 256$ places this pole near $Q_{t,1} = -0.00113463136 + 0.00667341664\,i$. It lies strictly inside $|\operatorname{Re} Q_j| < 0.06$, $|\operatorname{Im} Q_j| < 0.01$. Thus even a finite, exactly representable raw member violates holomorphy on the prescribed rectangle. This statement concerns the class, not the separately certified frozen checkpoint.

## *E.1 Radius peeling and the scope of a regularizer bound*

A countable radius decomposition can preserve the entire raw class without selecting a small neighborhood after fitting. The following statement is conditional on valid envelope and complexity bounds; no new numerical constants for them are asserted here.

**Proposition S9.** Conditional radius peeling. *Fix $N$, $\alpha \in (0,1)$ and an iid sampling law $P$ before drawing data; the joint sample law is $P^N$. For $j = 0, 1, \ldots$, put $R_j = 2^j$, $\Theta_j = \{\theta : \|\theta\|_\infty \le R_j\}$ and $\alpha_j = \alpha/[(j+1)(j+2)]$. Suppose $L_j$ is a valid uniform loss envelope on $\Theta_j$, and $\overline{\mathfrak{R}}_{N,j}$ is a valid bound for its empirical loss-class Rademacher complexity. With probability at least $1-\alpha$, simultaneously for every $j$ and every $\theta \in \Theta_j$,*

$$\mathbb{E}\ell_\theta \le \widehat{L}_N(\theta) + 2\overline{\mathfrak{R}}_{N,j} + 3L_j\sqrt{\frac{2\log(4/\alpha_j)}{N}}.$$

*The radius index may then be chosen from the fitted output.*

**Proof.** Apply Chen et al. (2022, Lemma 2) to each predeclared cube and use $\sum_{j\ge 0}\alpha_j = \alpha$. The cubes cover the finite-coordinate real parameter space and the probability event is simultaneous, so post-data radius selection does not remove coverage.

For a nonnegative regularized objective $F(\theta) \ge \lambda\|\theta\|_2^2/2$, a certified upper bound $F_0$ and exact monotone descent imply $\|\theta_t\|_\infty \le \sqrt{2F_0/\lambda}$. The recorded $\lambda = \mathrm{binary64}(10^{-10})$ and initial objective give a numerical radius near $18.16743$. A rigorous use needs a true-objective upper bound and verified descent; ordinary floating-point Armijo logs alone are insufficient. For example, if $F_0 \le 2\,10^{-8}$ and those descent premises were certified, radius 20 would suffice. Peeling can instead select a cube from exact stored final parameters without asserting optimizer premises. It still needs its stated class-uniform envelopes and complexity bounds, and does not guarantee a useful small-sample bound.

### E.2 A quantitative limit of the generic confidence term

The member in Proposition S7 with $t = 1$ lies in $\Theta_0$. At the interior point $q = (0, 3/64)$, outward evaluation of the exact stored reference gives

$$|\partial_2(V_1 - V_*)| > 8.24719\,10^{-12} =: b_*.$$

Hence every valid global envelope for a cube containing this member satisfies $L_{\max} \geq b_*^6$. For $\alpha = 0.05$ and $N = 40$, the sixth root of the confidence term alone is greater than $8.72730\,10^{-12}$. This exceeds the retained sufficient risk-root allowance $G = 1.0126174927\,10^{-13}$ by more than a factor of eighty. Even if the empirical loss and Rademacher term were set to zero, the displayed confidence bound could fit $G^6$ only if

$$N > 18\log(4/\alpha)(b_*/G)^{12}.$$

Using the unrounded interval endpoints gives $N > 6.7185\,10^{24}$. This is an obstruction to this particular conservative certificate, not an information-theoretic lower bound on learning or a proof that every statistical method needs that many observations. The bounded-cube envelopes above can be much looser still.

The fourteen training trajectories do not constitute iid phase-space samples and their profiled weak objective is not the empirical sixth-power gradient loss. The forty fresh configurations obey at most the declared independent finite-grid source model. Such a finite atomic law can enter the bounded-loss probability bound, but has no Lebesgue density; the Sobolev step requires the separately proved continuous-cell coupling and its class-uniform Lipschitz allowance. Rest-start trajectories plus independent momentum tags use the common-kinetic lifting identity rather than a claim that the physical initial conditions have a four-dimensional density.

These limitations do not invalidate the fixed-checkpoint output certificate or the corrected point-observation inverse certificate. They identify assumptions still needed for a simultaneous, whole-neural-class statistical learning claim. Failure of those assumptions for an unrestricted implementation is not a counterexample to the cited learning theorem.

## F Point-gradient observations and frozen-pair population risk

This section gives two complementary inputs to conservative certification. The first uses a realized finite point design and certified observation errors. The second evaluates the population risk directly for the known frozen reference pair. Neither replaces the weak-observation nuisance identity, the analytic remainder, or the independent KAM residual.

### F.1 Finite-dimensional excitation and a corrected left inverse

Let $\chi(q)$ be the same forty normalized monomials as in (S9), with $r_0 = \mathrm{binary64}(0.1)$ interpreted exactly. For configurations $Q_i$, retain both partial derivatives and define

$$A_{2i-1,j} = \partial_1\chi_j(Q_i), \qquad A_{2i,j} = \partial_2\chi_j(Q_i), \qquad i = 1, \ldots, 40.$$

Thus $A \in \mathbb{R}^{80\times 40}$. This is an observation matrix for the finite potential output, rather than a parametrization of the neural weights.

An exact excitation witness uses the Cartesian product of the twelve coordinates $\{-11, -9, \ldots, 9, 11\}/256$. If both gradient polynomials vanish at all 144 points, successive univariate interpolation makes them identically zero: each has degree at most eleven in either variable. The potential is consequently constant, hence zero in the stated degree-six-through-twelve space. Its $288 \times 40$ gradient matrix has rank forty; exact elimination selects a nonzero $40 \times 40$ minor. For its fixed derivative-coordinate selectors, the determinant at forty variable configurations is a nonzero polynomial of total degree

$$7 \cdot 5 + 9 \cdot 7 + 11 \cdot 9 + 13 \cdot 11 = 340.$$

Under independent uniform sampling of each configuration coordinate from the $2^{48}$ distinct centers in $[-1/16, 1/16]$, polynomial zero counting therefore gives rank failure probability at most $340/2^{48} < 1.208 \times 10^{-12}$. The full two-component matrix retains that selected minor. This probability assumes the independent uniform bit model; a realized operating-system

transcript does not prove physical-source independence. Rank alone supplies no inverse-gain bound. Dimension counting only gives the necessary condition $2N \geq 40$ for unrestricted exact full-gradient recovery, not the minimal sufficient $N$.

For the single realized forty-point design, all records and their original order are retained. Let $R \in \mathbb{R}^{40\times 80}$ be its saved dyadic approximate inverse, with quantum $2^{-160}$, and suppose

$$\eta = \|I - RA\|_\infty < 1, \qquad J = (RA)^{-1}R.$$

Then $JA = I_{40}$ and

$$\|J - R\|_\infty \leq \frac{\eta}{1-\eta}\,\|R\|_\infty\,. \tag{S78}$$

For this design the exact rational calculation gives $\eta < 1.410 \times 10^{-47}$. The approximate inverse is selected numerically; equation (S78) is checked with its saved integers and the exact realized matrix.

**Proposition S10.** Point-observation conservative error. *Write the same frozen error as $\widehat{V} - V_* = \chi^{\mathsf{T}} a_V + \rho_V$. Let $Y$ contain the two measured conservative-gradient components at each configuration, and set*

$$\xi = Y - \nabla V_*(Q), \quad z = \nabla\widehat{V}(Q) - Y, \quad t = \nabla\rho_V(Q),$$

*in the stated eighty-component order. Then*

$$z = Aa_V + t - \xi, \qquad a_V = Jz + J(\xi - t). \tag{S79}$$

*For a real or complex domain $\mathcal{O}$, suppose certified bounds satisfy*

$$B_\alpha \geq \sup_{\mathcal{O}} |(\partial^\alpha\chi)^{\mathsf{T}} Jz|, \qquad \beta_{\alpha,2} \geq \sup_{\mathcal{O}} \left\|(\partial^\alpha\chi)^{\mathsf{T}} J\right\|_2.$$

*If $\|\xi\|_2 \leq E$, $\|t\|_2 \leq T$ and $\|\partial^\alpha\rho_V\|_{\infty,\mathcal{O}} \leq R_\alpha$, then*

$$\left\|\partial^\alpha(\widehat{H} - H_*)\right\|_{\infty,\mathcal{O}} \leq B_\alpha + \beta_{\alpha,2}(E+T) + R_\alpha. \tag{S80}$$

**Proof.** Subtraction gives the first identity; multiplication by $J$ gives the second. Apply $\partial^\alpha\chi$ and Cauchy–Schwarz, then add the analytic remainder. The kinetic terms agree exactly.

Componentwise tail and noise bounds can be retained more sharply. If $g_j \geq \sup_{\mathcal{O}} |\partial^\alpha\chi_j|$, $t_i^+ \geq |t_i|$, and $u_j = \sum_i |R_{ji}| t_i^+$, equation (S78) gives the tail contribution

$$\sum_j g_j u_j + \left(\sum_j g_j\right)\frac{\eta}{1-\eta}\max_j u_j.$$

An analogous component-noise gain is defined by

$$\sup_{\mathcal{O}} \left\|(\partial^\alpha\chi)^{\mathsf{T}} J\right\|_1 \leq \sum_j g_j \left\|R_{j,:}\right\|_1 + \left(\sum_j g_j\right)\frac{\eta}{1-\eta}\,\|R\|_\infty =: \beta_{\alpha,\infty}.$$

For the complex rectangle $|\operatorname{Re} q_i| \leq 3/50$, $|\operatorname{Im} q_i| \leq 1/100$, a rational upper bound for $\sqrt{37/10000}$ controls each coordinate modulus. All derivatives through total order three, the point tails and final complex tails are included. Reserving half of each factor-512 partial-error budget for the nonzero signed principal $B_\alpha$ leaves a common sufficient component-noise allowance of $2.7262 \times 10^{-15}$. This predeclared sufficient condition is instantiated for the realized observations in Section F.3, retaining the separate signed principal and label-error terms.

### F.2 Rest-start observation labels and exact recorded clocks

The observation protocol sets $(q(0), p(0), u) = (Q, 0, 0)$. At that instant, the momentum-quadratic inertial term and the linear Rayleigh force vanish, so $p'(0) = -\nabla V_*(Q)$. At each stored node, its declared simulator observer exports the field-defined canonical velocity $v = G(q)^{-1}p$ in normalized time. Internal momentum is part of this simulator observer; it is not an independently measured physical sensor. Native momentum, stored time derivatives and force exports remain diagnostic files. The label construction reads only $q$, observed $v$, input and clock observations, reconstructing $\widetilde{\Pi} = G(q)v$ with the prescribed metric.

Let the actual exact nodes be $t_j = \mathrm{Q}(\omega_0)\,\mathrm{Q}(s_j)$, where $s_j$ are the stored binary64 SI times and Q denotes their exact rational interpretation. These differ conceptually from both rounded stored normalized times and nominal $jh$. For $0 = t_0 < t_1 < \cdots < t_4$, let $w_j$ be the derivative at zero of the corresponding Lagrange basis polynomial. Then

$$\sum_j w_j t_j^k = \begin{cases} 1, & k = 1, \\ 0, & k = 0, 2, 3, 4, \end{cases} \qquad Y = -\sum_{j=0}^{4} w_j \widetilde{\Pi}_j.$$

The scalar error functional $f'(0) - \sum_j w_j f(t_j)$ has the nonnegative Peano kernel

$$K(s) = -\frac{1}{24}\sum_j w_j (t_j - s)_+^4, \qquad \int_0^{t_4} K(s)\,ds = \frac{t_1 t_2 t_3 t_4}{120}.$$

For completeness, the repeated-node interpolation remainder has the sign of $f^{(5)}$; applying it to nonnegative continuous fifth derivatives proves kernel nonnegativity. The exact-node implementation also verifies nonnegative Bernstein coefficients of its four polynomial pieces. Thus a uniform Euclidean bound $\left\|p^{(5)}\right\|_2 \le M_5$ yields the vector truncation bound

$$M_5 \frac{t_1 t_2 t_3 t_4}{120}.$$

No additional dimension factor is needed for this Euclidean bound. For momentum-node error bounds $\|e_j\|_2 \le \varepsilon_j$, add $\sum_j |w_j|\varepsilon_j$ and the verified arithmetic allowance. The uniform rest-flow certificate permits nominal $h = 2^{-27}$; its fifth-derivative bound gives nominal truncation below $1.5470 \times 10^{-19}$. This term alone does not control native solver or observation error.

The nominal step is $h = 2^{-27}$ and the full normalized horizon is $2^{-25}$. With $\omega_0 \simeq 11.4172873\,\mathrm{s}^{-1}$, the latter corresponds to approximately $2.61 \times 10^{-9}\,\mathrm{s}$. This is a finite-Gauss simulation field-observer protocol designed for a strict truncation budget. Its forty configurations are the sampling units under the uniform-bit model; the 200 trajectory nodes do not constitute 200 iid samples. The Hamiltonian remains fixed, and these validation records are separate from both eighteen-trajectory training/reserved cohorts. Using a physical velocity sensor would require independent observation and clock error bounds as well as physical drag calibration. The specified viscosity is a numerical assumption, not a measured coefficient.

The evaluation procedure seals all forty observation-only labels and their source hashes before computing any reference-gradient discrepancy. The ensuing known-reference comparison encloses $\xi_i = Y_i - \nabla V_*(Q_i)$ at these realized points. Such a comparison does not establish a uniform observation-error law for an unknown plant.

### F.3 Acceptance for the realized forty-point observations

All forty rest-start trajectories produce five observation nodes under the same frozen checkpoint. The observation-only reconstruction is sealed before controlled-reference discrepancies are computed. The resulting maximum component error is at most $7.850603 \times 10^{-16}$, and the entire eighty-component error vector has Euclidean norm at most $2.040666 \times 10^{-15}$. These bounds satisfy the predeclared strict inverse's component and vector capacities at the realized configurations.

The observed principal remains independent of the reference-gradient comparison: $z = \nabla\widehat{V}(Q) - Y$ is formed from the frozen learned output and sealed labels alone. Let $b_R = Rz$ and let $\xi_i^+$ enclose the absolute controlled-reference label error.

For the derivative weights $g_j$ defined above, the evaluated terms are

$$B_\alpha = \sum_j g_j |b_{R,j}|^+ + \left(\sum_j g_j\right) \frac{\eta}{1-\eta} \|b_R\|_\infty^+ ,$$
$$u_{\xi,j} = \sum_i |R_{ji}| \xi_i^+ ,$$
$$N_{\alpha,\Gamma} = \sum_j g_j u_{\xi,j} + \left(\sum_j g_j\right) \frac{\eta}{1-\eta} \max_j u_{\xi,j},$$
$$N_\alpha = \min\{N_{\alpha,\Gamma}, \beta_{\alpha,2} \|\xi\|_2^+\}.$$

The $J-R$ correction follows from $Jz - b_R = [(RA)^{-1} - I]b_R$. The two noise bounds are separately valid, so their minimum is valid. Table S3 retains the observed principal, selected noise, point-tail transport $T_\alpha^{\mathrm{pt}}$ and final complex remainder $R_\alpha$. Each total is computed from exact rational inputs before its displayed upper rounding. All nine signed-principal reservations and all nine complete partial-error budgets satisfy the predeclared factor-512 conditions.

**Table S3.** Realized point-observation derivative bounds on the complex rectangle. Every numerical entry is rounded upward; totals use unrounded rational arithmetic. $T_\alpha^{\mathrm{pt}}$ retains the point-observation tail, and $R_\alpha$ the independent final analytic remainder.

| $\alpha$ | $B_\alpha$ | $N_\alpha$ | $T_\alpha^{\mathrm{pt}}$ | $R_\alpha$ | Total |
|---|---|---|---|---|---|
| $(0,1)$ | $1.0960\times10^{-11}$ | $5.5843\times10^{-11}$ | $1.2764\times10^{-10}$ | $1.2440\times10^{-14}$ | $1.9445\times10^{-10}$ |
| $(1,0)$ | $8.8484\times10^{-12}$ | $4.7289\times10^{-11}$ | $1.4637\times10^{-10}$ | $1.2440\times10^{-14}$ | $2.0252\times10^{-10}$ |
| $(0,2)$ | $1.0616\times10^{-9}$ | $5.4204\times10^{-9}$ | $1.0968\times10^{-8}$ | $1.5999\times10^{-12}$ | $1.7451\times10^{-8}$ |
| $(1,1)$ | $6.3771\times10^{-10}$ | $3.1221\times10^{-9}$ | $8.5380\times10^{-9}$ | $1.5999\times10^{-12}$ | $1.2300\times10^{-8}$ |
| $(2,0)$ | $6.9403\times10^{-10}$ | $4.1137\times10^{-9}$ | $1.3799\times10^{-8}$ | $1.5999\times10^{-12}$ | $1.8608\times10^{-8}$ |
| $(0,3)$ | $1.0311\times10^{-7}$ | $5.3075\times10^{-7}$ | $9.8113\times10^{-7}$ | $2.0248\times10^{-10}$ | $1.6152\times10^{-6}$ |
| $(1,2)$ | $4.9646\times10^{-8}$ | $2.3404\times10^{-7}$ | $5.6766\times10^{-7}$ | $2.0248\times10^{-10}$ | $8.5154\times10^{-7}$ |
| $(2,1)$ | $3.9930\times10^{-8}$ | $2.0490\times10^{-7}$ | $6.3123\times10^{-7}$ | $2.0248\times10^{-10}$ | $8.7626\times10^{-7}$ |
| $(3,0)$ | $5.4920\times10^{-8}$ | $3.7597\times10^{-7}$ | $1.3100\times10^{-6}$ | $2.0248\times10^{-10}$ | $1.7410\times10^{-6}$ |

The orderwise maximum totals are at most $2.025150\times10^{-10}$, $1.860781\times10^{-8}$ and $1.740994\times10^{-6}$. Their ordered field increments use the unchanged factors $(1,2,4,4)$. The recomputed same-$K_0$ torsion Neumann quantity is below $0.02996176$, and its same-cubic-response counterpart is below $0.07143854$. All eight canonical KAM gates are below one, with largest gate below $2.205303\times10^{-4}$. The complete invariance residual, assembled from the retained Fourier interpolants, alias tails and recomputed parameter tail, is below $1.205897\times10^{-41}$. The corrected-embedding displacement is below $1.614613\times10^{-23}$.

These are deterministic finite-output conclusions using the realized point design, frozen-output analytic remainders and controlled-reference fidelity. The independently uniform bit law remains an explicit sampling premise. The eighty derivative components do not supply a continuous-density, class-uniform neural learning theorem. Neither this acceptance nor the oracle calculation below identifies an unknown physical plant or changes the learned Hamiltonian.

### *F.4 Oracle verification of the frozen population-risk premise*

For the same frozen pair, write $P = \chi^{\mathsf{T}} a_V$ and let $\tau_i$ bound $\partial_i \rho_V$ on $[-r,r]^2$, $r = 1/16$. All real norms in this subsection use the normalized uniform probability measure $dq/(2r)^2$. The identical kinetic energies make the momentum components of the Hamiltonian-gradient error zero, so an independent known-kinetic momentum lift leaves this risk unchanged. For $x_i = q_i/r_0$, its exact moments are

$$\mathbb{E}(x_1^a x_2^b) = \begin{cases} (r/r_0)^{a+b}/[(a+1)(b+1)], & a,b \text{ even}, \\ 0, & \text{otherwise}. \end{cases} \tag{S81}$$

The coefficient of $x_1^{a-1}x_2^b$ in $\partial_1 P$ is $a(a_V)_{ab}/r_0$, with the analogous factor $b/r_0$ in the second component. Outward polynomial convolution computes $I_i = \mathbb{E}[(\partial_i P)^6]$. Since these are real polynomials, the sixth power equals the absolute sixth power.

**Proposition S11.** Frozen-output risk-to-derivative verification. *If* $u_i^6 \geq I_i$ *and*

$$R_6^6 \geq (u_1 + \tau_1)^6 + (u_2 + \tau_2)^6,$$

*then*

$$\left( \mathbb{E} \sum_i |\partial_i (\widehat{V} - V_*)|^6 \right)^{1/6} \leq R_6.$$

*Let* $K_j$ *be the previously verified normalized-*$L^2$ *polynomial gains to complex partials of orders* $j = 0, 1, 2$*, and let* $T_k$ *bound the complex analytic remainder partials of total order* $k$*. Then*

$$\sup_{\mathcal{O}} |\partial^\alpha (\widehat{V} - V_*)| \leq K_{k-1}(R_6 + \max_i \tau_i) + T_k, \qquad |\alpha| = k \in \{1, 2, 3\}. \tag{S82}$$

**Proof.** Minkowski's inequality bounds each scalar component in $L^6$ by $u_i + \tau_i$. Sum their sixth powers. On a probability space, $\left\| \partial_i (\widehat{V} - V_*) \right\|_2 \leq R_6$, whence $\|\partial_i P\|_2 \leq R_6 + \tau_i$. Apply the polynomial gain to a suitable gradient component and its derivatives, then add the complex remainder.

The actual computation uses independently enclosed Taylor coefficients, Arb arithmetic at 512 bits, and exact rational moments. Sixth-root upper bounds are obtained by integer arithmetic with denominator $2^{384}$, checking $(k-1)^6/2^{2304} < y \leq k^6/2^{2304}$. The analytic remainder is retained throughout; reuse of the older arbitrary-coordinate allowance only enlarges this bound. The resulting risk root is below $2.395279 \times 10^{-14}$. Formula (S82) gives order-one, order-two and order-three partial bounds below

$$1.483543 \times 10^{-10}, \qquad 4.015155 \times 10^{-8}, \qquad 8.988099 \times 10^{-6},$$

respectively. They satisfy the predeclared factor-512 ordered derivative budget. Recomputing the same-pair Gram, mean torsion, parameter tail and eight canonical KAM gates gives a largest gate below $2.385535 \times 10^{-4}$. The complete embedding residual remains below $1.205897 \times 10^{-41}$ and is assembled separately from this risk.

This verifies a population-risk premise by access to the known reference function. It does not infer that risk from COMSOL observations, validate an empirical complexity term, or instantiate the class-uniform statistical theorem. The same frozen endpoints, frequency, cubic response, low-Fourier interpolants, large-domain analytic tail and embedding-tube premises are retained. New fitted endpoints would require their own certificates.

## G Simultaneous acceptance of data-adaptive conservative outputs

A point-observation certificate can validate outputs chosen using the same observations. The relevant uniformity is a deterministic implication for every output satisfying its own verified remainder conditions. It does not require a common loss envelope for all raw network parameters. In particular, the unbounded-class obstructions do not invalidate the result below, and the result does not remove those obstructions from a population-risk theorem.

### *G.1 A common design and output-specific remainder certificates*

The reference and candidate Hamiltonians have the same known kinetic energy,

$$H_*(q,p) = \tfrac{1}{2} p^T G(q)^{-1} p + V_*(q), \qquad H_\theta(q,p) = \tfrac{1}{2} p^T G(q)^{-1} p + V_\theta(q).$$

The target $V_*$ is fixed before sampling. Let $V_4$ denote the common known potential through degree four, and let $\chi$ contain the forty monomials $(q_1/r_0)^a (q_2/r_0)^b$ of total degrees $6, 8, 10, 12$, where $r_0 = \mathrm{binary64}(0.1)$ is interpreted exactly. For an admissible output, its certificate establishes decompositions

$$V_* = V_4 + \chi^T c_* + r_*, \qquad V_\theta = V_4 + \chi^T c_\theta + r_\theta. \tag{S83}$$

Set $a_\theta = c_\theta - c_*$ and $\rho_\theta = r_\theta - r_*$. A verified target remainder can be shared; the candidate remainder must belong to the actual deployed output. Neither the unknown coefficients $c_*$ nor a bound on the full error $V_\theta - V_*$ is assumed known.

Retain both gradient components at configurations $Q_i$, in configuration-major order. With $m = 2N$, fix the observed labels $Y$ and the common design

$$A_{(i,j),k} = \partial_j \chi_k(Q_i), \qquad Y_{ij} = \partial_j V_*(Q_i) + \xi_{ij}, \quad j = 1, 2. \tag{S84}$$

The exact rational interpretation of configurations and normalization is part of the design. The label-error contract is a single componentwise bound $|\xi| \le b^\xi$ for these same labels; it is not chosen separately to favor each candidate.

Let $\mathcal{Q}_{\mathrm{obs}}$ contain the observed configurations, and let $\mathcal{O} \subset \mathbb{C}^2$ be the declared complex domain required downstream. They need not be the same domain. The reference and every admitted output are real-valued on the real observation domain. They are differentiable on a neighborhood of $\mathcal{Q}_{\mathrm{obs}}$ and holomorphic on a neighborhood of $\mathcal{O}$. For each admitted output, its remainder certificate supplies

$$|\partial_j \rho_\theta(Q_i)| \le b^{t,\theta}_{ij}, \qquad \|\partial^\alpha \rho_\theta\|_{\infty,\mathcal{O}} \le T_{\alpha,\theta}. \tag{S85}$$

The observation-point bounds cannot be inferred from a complex-domain bound if some $Q_i$ lie outside that domain. Different outputs may have different valid bounds or fail to obtain such a certificate altogether.

Choose a rational $R \in \mathbb{Q}^{40\times m}$, and validate

$$\|I - RA\|_\infty \le \eta < 1, \qquad J = (RA)^{-1}R.$$

Here $R$ can be computed after observing the design. This is a deterministic matrix certificate, not an assumption about its numerical condition number. Let $u_{\alpha,k} \ge \sup_{\mathcal{O}} |\partial^\alpha \chi_k|$, $U_\alpha = \sum_k u_{\alpha,k}$, and define, for $b \ge 0$,

$$\Gamma_\alpha(b) = \sum_k u_{\alpha,k}(|R|b)_k + U_\alpha \frac{\eta}{1-\eta} \||R|b\|_\infty. \tag{S86}$$

**Theorem S5.** Simultaneous certified output acceptance. *Fix the design, labels and a valid common label-error contract in* (S84)*. Suppose the matrix certificate above holds. For every candidate $\theta$ having sound certificates* (S83)–(S85)*, define*

$$z_{\theta,ij} = \partial_j V_\theta(Q_i) - Y_{ij}.$$

*Suppose validated evaluation and polynomial arithmetic provide*

$$Z_\theta \ge \|Rz_\theta\|_\infty, \qquad P_{\alpha,\theta} \ge \sup_{\mathcal{O}} |(\partial^\alpha \chi)^T R z_\theta|,$$

*including all evaluation and label-arithmetic enclosure errors. Set*

$$\begin{aligned} B_{\alpha,\theta} &= P_{\alpha,\theta} + U_\alpha \frac{\eta}{1-\eta} Z_\theta, \\ \mathcal{E}_{\alpha,\theta} &= B_{\alpha,\theta} + \Gamma_\alpha(b^\xi) + \Gamma_\alpha(b^{t,\theta}) + T_{\alpha,\theta}. \end{aligned} \tag{S87}$$

*Then, simultaneously for every such candidate,*

$$\|\partial^\alpha(V_\theta - V_*)\|_{\infty,\mathcal{O}} \le \mathcal{E}_{\alpha,\theta}. \tag{S88}$$

*In particular, a rule accepting exactly those candidates whose verified $\mathcal{E}_{\alpha,\theta}$ satisfy declared derivative budgets is sound for all its accepted outputs. Candidates may be trained, selected, or repeatedly revised using the same $Q, Y$, and their number need not be fixed in advance. No parameter-selection probability penalty is required for this deterministic implication.*

**Proof.** For every candidate separately, differentiation of the exact decompositions gives the signed identities

$$z_\theta = Aa_\theta + t_\theta - \xi, \qquad a_\theta = Jz_\theta + J(\xi - t_\theta), \qquad JA = I_{40}.$$

Write $E = I - RA$. The Neumann series proves

$$\|(RA)^{-1} - I\|_\infty \le \frac{\eta}{1-\eta}.$$

Consequently the observable coefficient term satisfies

$$\sup_{\mathcal{O}} |(\partial^\alpha \chi)^T J z_\theta| \le P_{\alpha,\theta} + U_\alpha \frac{\eta}{1-\eta} Z_\theta.$$

For every $w$ with $|w| \le b$, the same series gives $\sup_{\mathcal{O}} |(\partial^\alpha \chi)^T J w| \le \Gamma_\alpha(b)$. Apply this inequality to $\xi$ and $-t_\theta$, and add the derivative of $\rho_\theta$. This proves (S88). Every step is an algebraic or certified analytic implication on the same data realization. Since $\theta$ was arbitrary and no step invoked a fixed-output probability statement, the implications hold simultaneously. Selection after observing the data does not change them.

The theorem retains nonzero measured residuals: neither interpolation of the labels nor an optimizer-convergence assumption appears in (S87). Signed coefficients in $Rz_\theta$ can be preserved in the polynomial bound $P_{\alpha,\theta}$. Bounds on individual coefficient magnitudes are also valid, but can be weaker. An independently validated gain $\beta_\alpha \ge \sup_{\mathcal{O}} \|(\partial^\alpha \chi)^T J\|_2$ allows the noise term to be replaced by $\min\{\Gamma_\alpha(b^\xi), \beta_\alpha E_\xi\}$ if the same label contract also proves $\|\xi\|_2 \le E_\xi$.

The residual in this theorem belongs to the deployed potential. If a training nuisance coefficient $h$ is instead added before profiling, the identity recovers $a_\theta + h$; the deployed coefficient then contains the additional term $-h$. A small profiled residual does not authorize omitting that term. Direct evaluation of $z_\theta$ as defined above avoids that ambiguity.

## G.2 The probability quantifier and its limits

For the prospectively declared one-draw design, each of the eighty configuration coordinates is independent and uniform on the 48-bit center grid in $[-1/16, 1/16]$. The fixed-coordinate determinant witness has total degree 340 and is not the zero polynomial. Therefore the full 80-by-40 design satisfies

$$\mathbb{P}\{\operatorname{rank} A < 40\} \le \frac{340}{2^{48}} < 1.208 \times 10^{-12}. \tag{S89}$$

This event depends on the design, not on a subsequently selected network. On the full-rank event a real left inverse exists; density of the rationals also gives a rational $R$ with $\eta < 1$. This existence claim does not bound its gain or guarantee that a particular numerical procedure finds it.

If a common label-error event $\mathsf{E}_\xi$ has probability at least $1 - \delta_\xi$ under the declared joint observation law, then, with probability at least $1 - 340/2^{48} - \delta_\xi$, the design is full rank and the preceding simultaneous implication applies to every output with sound remainder and arithmetic certificates. Independence of $\mathsf{E}_\xi$ and the design is unnecessary for this union bound. The event must cover all retained labels jointly and cannot be an output-by-output assertion reused after selection. A change of observer that changes $Y$ requires the corresponding new label contract.

A successful deterministic matrix and label certificate already proves the implication for its realized data; it does not become more valid because the rank-failure bound is small. Equation (S89) describes excitation of a new draw, not the probability of good conditioning, accurate observations, successful fitting, small tails, or KAM acceptance. Failed candidates are not asserted to satisfy any common derivative budget. Conditioning screens, redraws, point deletion and a changed feature space require their actual sampling and algebraic contracts to be stated. The independent-uniform-bits model is a probability premise, not a theorem about a physical random-number source. No continuous-density claim is made for the finite center grid.

## G.3 From the accepted function bound to a candidate KAM interface

Let $\mathcal{E}_{k,\theta}$ be the maximum of (S87) over multi-indices of total order $k$. On a phase domain where the common kinetic energy is defined, the potential-only difference and the canonical norm conventions give

$$(\delta X_0, \delta X_1, \delta X_{t1}, \delta X_2) \le (\mathcal{E}_{1,\theta},\, 2\mathcal{E}_{2,\theta},\, 4\mathcal{E}_{2,\theta},\, 4\mathcal{E}_{3,\theta}). \tag{S90}$$

The factors count ordered derivative entries; the two second-order quantities use different operator and ordered-derivative norms. This is a computable data-to-function and data-to-field conclusion before any torus assertion is made.

For a chosen candidate, these increments can enter a KAM verification with an explicitly bound approximate embedding, frequency and response. That verification must still calculate the candidate's complete invariance defect, including low Fourier coefficients and independent analytic Fourier and parameter tails, and check the open domain, arithmetic, embedding geometry, torsion inverse and scalar correction inequalities. Those are independently checkable witness conditions, not consequences of fitting the observations. A new candidate cannot inherit the old cubic response or its low-Fourier residual merely by passing (S87). The same observation design may be reused, whereas every model-dependent downstream witness must remain bound to the candidate it verifies.

### *G.4 Instantiation for the presently accepted output*

The retained experiment supplies forty configurations and eighty gradient labels from prospectively declared rest-start observations. All forty records are retained. The accepted matrix certificate has $\eta < 1.409545 \times 10^{-47}$. The observation box has halfwidth $1/16$; the derivative bounds below concern the separate rectangle

$$|\operatorname{Re} q_j| \le 3/50, \qquad |\operatorname{Im} q_j| \le 1/100, \qquad j = 1, 2.$$

For the current deployed output, the sealed-label comparison gives

$$\max_{i,j} |\xi_{ij}| < 7.850603 \times 10^{-16}, \qquad \|\xi\|_2 < 2.040666 \times 10^{-15}.$$

These are controlled-reference fidelity bounds for the realized observations. They are neither an unknown-plant noise certificate nor a uniform continuous-flow observation-error theorem. The measured principal term is nonzero; its order-one bound is below $1.095927 \times 10^{-11}$. The separately validated observation-point and complex-domain remainders are retained in the totals in Table S4. The displayed numbers are outward decimal summaries of the exact rational certificate.

**Table S4.** Maximum derivative error over each total order, obtained from the measured principal, actual label-error gain, point-tail gain and complex remainder for the current output.

| Derivative order $k$ | Certified error below | Declared budget above |
|---|---|---|
| 1 | $2.025150 \times 10^{-10}$ | $4.261915 \times 10^{-9}$ |
| 2 | $1.860781 \times 10^{-8}$ | $2.244928 \times 10^{-7}$ |
| 3 | $1.740994 \times 10^{-6}$ | $2.591881 \times 10^{-5}$ |

The corresponding ordered increments have also been substituted into the independently retained same-pair KAM calculation. Its complete embedding residual is below $1.205897 \times 10^{-41}$, and the largest of its eight dimensionless scalar acceptance ratios is below $2.205303 \times 10^{-4}$. This is one accepted instance of the theorem; the forty observations were not used to retrain this frozen output. The simultaneous theorem additionally permits future data-adaptive outputs to be tested against the same labels, without asserting that any such future output has already been trained or accepted.

The conclusion concerns validated conservative outputs with a common known kinetic energy and finite polynomial representative. It does not give a class-uniform population-risk bound, instantiate the continuous-density statistical-learning theorem, establish complex network superiority, or guarantee identification of an unknown physical plant.

## H A full-phase extension and its verified error interface

The following results retain the generic dependence on both canonical position and momentum. They concern the precision-centered GC output, which is distinct from the structured Hamiltonian certified in the article. Its configuration slice is observed at rest, while its momentum derivatives require separate analytic bounds. The common kinetic energy cancels in the Hamiltonian difference; this cancellation does not identify the physical velocity with the new learned Hamiltonian's momentum derivative.

### H.1 Transport from rest observations to a full-phase Hamiltonian

Consider the reference and a fixed candidate

$$H_*(q,p) = T(q,p) + V_4(q) + U(q), \qquad \widehat{H}(q,p) = T(q,p) + V_4(q) + N(q,p), \tag{S91}$$

where $q, p \in \mathbb{R}^2$ are the physical canonical coordinates and $T(q,p) = \frac{1}{2} p^T G(q)^{-1} p$ is exactly the same known kinetic energy in both models. The full-phase correction $N$ is not replaced by its restriction to $p = 0$. Define

$$E = \widehat{H} - H_* = A + B, \qquad A(q) = N(q,0) - U(q), \qquad B(q,p) = N(q,p) - N(q,0). \tag{S92}$$

Any discrepancy in the kinetic term or in $V_4$ would require an additional error term; it is not included in (S92).

*H.1.1 A momentum-fiber transport bound* Let $\mathcal{D} \subset \mathbb{C}_q^2 \times \mathbb{C}_p^2$ be compact, let $\mathcal{Q}$ be its configuration projection, and put

$$\mathcal{D}^\star = \{(q, tp) : (q,p) \in \mathcal{D},\ 0 \le t \le 1\}, \qquad r_{p,j} = \sup_{(q,p)\in\mathcal{D}} |p_j|.$$

Suppose $N$ is holomorphic on a neighborhood of $\mathcal{D}^\star$ and $U$ on a neighborhood of $\mathcal{Q}$. These are holomorphic extensions in the four independent complexified canonical coordinates. They do not assert holomorphic dependence of a real-valued energy on the physical variable $(q + ip)/\sqrt{2}$ alone. For multi-indices $\alpha, \beta \in \mathbb{N}_0^2$, let

$$M_{\alpha,\beta} \ge \sup_{\mathcal{D}^\star} |\partial_q^\alpha \partial_p^\beta N|, \qquad e_\alpha \ge \sup_{\mathcal{Q}} |\partial_q^\alpha A|.$$

All derivatives here are in physical coordinates. Input normalization of a network must therefore be included in these bounds.

**Lemma S3.** Slice and momentum variation. *Under these premises, for every supplied multi-index,*

$$\|\partial_q^\alpha E\|_{\infty,\mathcal{D}} \le e_\alpha + \sum_{j=1}^{2} r_{p,j} M_{\alpha,e_j}, \tag{S93}$$

$$\|\partial_q^\alpha \partial_p^\beta E\|_{\infty,\mathcal{D}} \le M_{\alpha,\beta}, \qquad |\beta| \ge 1. \tag{S94}$$

*Here $e_j$ denotes the $j$th coordinate multi-index in momentum space. In particular, transporting a third pure configuration derivative by (S93) requires a fourth-order derivative of $N$.*

**Proof.** The segment $(q, tp)$ lies in $\mathcal{D}^\star$. The fundamental theorem of calculus, differentiated with respect to $q$, gives

$$\partial_q^\alpha B(q,p) = \int_0^1 \sum_{j=1}^{2} p_j \partial_q^\alpha \partial_{p_j} N(q, tp)\, dt.$$

Its absolute value is bounded by the sum in (S93). Adding the derivative of $A$ proves that inequality. If $|\beta| \ge 1$, the terms $U(q)$ and $N(q,0)$ differentiate to zero, so the mixed derivative of $E$ is exactly that of $N$. This proves (S94).

If the declared domain is star-shaped in each momentum fiber about zero, then $\mathcal{D}^\star \subset \mathcal{D}$. A tube around a nonzero-momentum invariant torus need not have this property. In that case the analyticity and derivative certificates must cover the enlarged fiber hull, even when the downstream dynamics use only the original tube. A real-box first- or second-derivative certificate alone does not establish this complex-domain premise or the required fourth-order bound.

Total evenness $N(-q,-p) = N(q,p)$ does not make $B$ quadratic in $p$. For example, $N(q,p) = \varepsilon q_1 p_1$ is totally even and vanishes identically on $p = 0$, together with its configuration gradient there, but $N_{p_1}(q,0) = \varepsilon q_1$. Thus rest configuration-gradient labels do not identify the momentum derivatives. This example also has $N_{pp} = 0$: positive definiteness of $\widehat{H}_{pp}$ does not imply $\widehat{H}_p = G^{-1}p$. A quadratic-in-momentum transport estimate would require the additional identity $N_p(q,0) = 0$ throughout the configuration domain, rather than total evenness alone.

*H.1.2 A conditional observation operator for the configuration slice* Suppose $A$ is even, $A(0) = 0$, and has a certified decomposition

$$A(q) = \chi_{48}(q)^T a + \rho(q), \qquad \chi_{48} = \big((q_1/r_0)^a (q_2/r_0)^b\big)_{a+b\in\{2,4,6,8,10,12\}}, \tag{S95}$$

where $r_0 = \mathrm{binary64}(0.1)$ is interpreted exactly. There are $3+5+7+9+11+13 = 48$ columns. The degree-two and degree-four columns are necessary because a generic full-phase correction is not constrained to start at degree six. If either the evenness or the gauge premise is absent, the feature space must be adjusted instead of assuming (S95).

At forty real rest configurations, retain both configuration-gradient components in configuration-major order. Let

$$\begin{aligned} \mathsf{A}_{(i,j),k} &= \partial_{q_j}\chi_{48,k}(Q_i), & Y_{ij} &= \partial_{q_j}(V_4+U)(Q_i) + \xi_{ij}, \\ z_{ij} &= \partial_{q_j}\widehat{H}(Q_i,0) - Y_{ij}, & t_{ij} &= \partial_{q_j}\rho(Q_i). \end{aligned} \tag{S96}$$

Since $T_q(q,0) = 0$, the exact signed identity is

$$z = \mathsf{A}a + t - \xi. \tag{S97}$$

This identity uses the same eighty labels for every candidate. It does not treat the two components or the time nodes used to construct a label as additional independent initial conditions.

The following is an interface conditional on a new matrix certificate. An existing eighty-by-forty certificate does not certify the present eighty-by-forty-eight matrix. Assume a rational $R \in \mathbb{Q}^{48\times 80}$ satisfies

$$\|I - R\mathsf{A}\|_\infty \le \eta < 1, \qquad J = (R\mathsf{A})^{-1}R, \qquad J\mathsf{A} = I_{48}.$$

For each derivative set $u_{\alpha,k} \ge \sup_{\mathcal{Q}} |\partial^\alpha \chi_{48,k}|$ and $U_\alpha = \sum_k u_{\alpha,k}$. For a nonnegative eighty-vector $b$, define

$$\Gamma_\alpha(b) = \sum_k u_{\alpha,k}(|R|b)_k + U_\alpha \frac{\eta}{1-\eta}\,\||R|b\|_\infty. \tag{S98}$$

Assume a joint contract $|\xi| \le b^\xi$, point-tail bounds $|t| \le b^t$, and a final-domain remainder bound $\|\partial^\alpha\rho\|_{\infty,\mathcal{Q}} \le T_\alpha$. The point-tail contract must cover the actual observed configurations, including any outside $\mathcal{Q}$. If validated arithmetic supplies

$$Z \ge \|Rz\|_\infty, \qquad P_\alpha \ge \sup_{\mathcal{Q}} |(\partial^\alpha\chi_{48})^T Rz|,$$

then a valid slice bound is

$$e_\alpha = P_\alpha + U_\alpha \frac{\eta}{1-\eta} Z + \Gamma_\alpha(b^\xi) + \Gamma_\alpha(b^t) + T_\alpha. \tag{S99}$$

Indeed, (S97) gives $a = Jz + J\xi - Jt$, and the Neumann estimate $\|(R\mathsf{A})^{-1} - I\|_\infty \le \eta/(1-\eta)$ proves (S99). Thus both the signed $Rz$ term and the $J - R$ correction are retained. A certified gain $\beta_\alpha \ge \sup_{\mathcal{Q}} \|(\partial^\alpha\chi_{48})^T J\|_2$ allows the noise term to be replaced by $\min\{\Gamma_\alpha(b^\xi), \beta_\alpha E_\xi\}$ only when the same label contract proves $\|\xi\|_2 \le E_\xi$. All network-evaluation, label-arithmetic and polynomial-enclosure errors must enter the stated bounds. An ordinary floating-point gradient is not by itself an exact $z$ certificate.

This is the finite-output acceptance algebra with a different feature space; it neither identifies raw network weights nor supplies a new matrix certificate. Sound candidate-specific tails and a valid common label contract allow the deterministic implication to hold for all accepted data-adaptive outputs on the same data realization. No population-risk statement or sampling probability is used here.

*H.1.3 Full-dimensional canonical derivative budgets* Write $x = (q_1, q_2, p_1, p_2)$ and let $C_\nu$ be the bounds obtained from (S93)–(S94) for every $\nu \in \mathbb{N}_0^4$ with $1 \le |\nu| \le 3$. For $\delta X = J_4 \nabla E$, where $J_4 = \left(\begin{smallmatrix} 0 & I_2 \\ -I_2 & 0 \end{smallmatrix}\right)$, the canonical matrix is a signed

permutation. With the unweighted coordinate maximum norm, valid budgets are

$$\begin{aligned}
\|\delta X\|_\infty &\le \Delta_0 := \max_a C_{e_a},\\
\|D\delta X\|_{\infty\to\infty} &\le \Delta_1 := \max_a \sum_b C_{e_a+e_b},\\
\sum_{a,b} |(D\delta X)_{ab}| &\le \Delta_{\Sigma,1} := \sum_{a,b} C_{e_a+e_b},\\
\|D^2\delta X\|_{\infty,\infty\to\infty} &\le \Delta_2 := \max_a \sum_{b,c} C_{e_a+e_b+e_c}.
\end{aligned} \tag{S100}$$

All coordinate indices in (S100) run from one to four, and the last sum includes ordered pairs. Consequently, if $b_k = \max_{|\nu|=k} C_\nu$, sufficient scalar bounds are

$$(\Delta_0, \Delta_1, \Delta_{\Sigma,1}, \Delta_2) \le (b_1, 4b_2, 16b_2, 16b_3). \tag{S101}$$

The potential-only factors $(1, 2, 4, 4)$ cannot be transferred to a full-phase correction. The entry-sum bound also dominates either induced matrix norm and can therefore bound a transposed-Jacobian quantity when that verifier uses the corresponding convention. Weighted norms or differently normalized tensors require their explicit coordinate factors instead.

The resulting budgets form an input to a candidate-specific KAM verification, not its conclusion. That verification must still bind the actual full Hamiltonian, complex domain and momentum-fiber hull, embedding and frequency, invariance residual, response operators, geometry, torsion and scalar smallness gates. An older embedding may be used as a numerical guess; an older same-pair certificate is not inherited by a new full-phase network. Likewise, forty rest labels do not identify its momentum dependence, and this transport argument neither changes the model to a configuration-only network nor proves a general statistical generalization theorem or a control guarantee.

### *H.2 Verified instance for the precision-centered GC output*

The selected GC output at update 385 has the actual form (S91), with $T$ given by the same direct eight-point-per-element Gauss graph and $U = V_* - V_4$. Its raw binary64 parameters and normalization buffers are fixed. The layer constraints are recomputed with interval arithmetic from those raw values, including the source's pre-rounded smooth-absolute scalar, rather than from exported floating-point effective weights.

For the independent-coordinate complexification of Section H.1, consider the common physical domain

$$\mathcal{D} = \{x \in \mathbb{C}^4 : |\operatorname{Re} x_j| \le 3/50, \quad |\operatorname{Im} x_j| \le 1/100,\ j = 1, \dots, 4\}. \tag{S102}$$

It contains the full momentum-fiber hull used in Lemma S3. The maximum imaginary preactivation is bounded above by $0.414019$ in the first layer and $0.828110$ in the second, both strictly below $\pi/2$. Thus both literal tanh layers are pole-free on a neighborhood of this compact domain. Positive four-variable Taylor majorants, composed through the two actual layers, bound all 69 nonzero mixed derivatives through total order four. Define $m_k = \max_{|\nu|=k} \|\partial^\nu N\|_{\infty,\mathcal{D}}$. Rounded upward for display,

$$\begin{aligned}
m_1 &\le 1.762215 \times 10^{-8}, & m_2 &\le 9.773950 \times 10^{-7},\\
m_3 &\le 1.077607 \times 10^{-4}, & m_4 &\le 1.725501 \times 10^{-2}.
\end{aligned}$$

Input scales, coefficient conjugation, central evenization and the energy gauge are included. A coefficient at a four-variable monomial $h^\nu$ is converted to an ordinary derivative using the multi-index factorial $\nu!$, not $|\nu|!$.

The forty rest observations supply an exact eighty-by-forty-eight gradient design for (S95). Its validated Neumann defect is less than $2.976200 \times 10^{-10}$. The principal uses strict evaluation of $\widehat{H}_q(Q_i, 0) - Y_i$, with the actual $V_{4,q}$ and all eighty sealed label components. No reference gradient is substituted for $Y_i$. The same controlled-reference label-fidelity contract is transported through the componentwise and vector-$L^2$ inverse gains; it is not a physical-sensor error theorem.

The original rectangular Cauchy tests at radii $0.065, 0.07, 0.08, 0.10, 0.12, 0.15, 0.20$ do not certify the required pole separation. No successful Cauchy premise is inferred from them. Instead, let $P_N$ and $P_U$ be the respective finite even Taylor polynomials through degree twelve, and set $t_N = \nabla(N(\cdot,0) - P_N)(Q_i)$ and $t_U = \nabla(U - P_U)(Q_i)$. The signed data identity becomes

$$z = \mathsf{A}a + t_N - t_U - \xi, \qquad a = J(z - t_N) + J(t_U + \xi).$$

The signed interval $t_N$ is subtracted before applying the inverse. The final candidate remainder uses the independently valid inequality

$$\|\partial^\alpha(N(\cdot,0) - P_N)\|_{\infty,\mathcal{Q}} \le M_{\alpha,0} + \sum_k u_{\alpha,k}|(P_N)_k|.$$

This is a finite-polynomial triangle bound and does not assert convergence of the origin Taylor series on a larger disk. The $J - R$ correction, common label error, reference point-tail gain and both final remainders are retained. Observation-point remainders cover the actual forty configurations, including those outside the final configuration rectangle. Table S5 gives the resulting complete slice bounds, rounded upward.

**Table S5.** Complete bounds for $\|\partial_q^\alpha[N(q,0) - U(q)]\|_{\infty,\mathcal{Q}}$, where $\mathcal{Q}$ is the projection of (S102). The bounds include the observation and analytic-tail terms.

| $\alpha$ | Upper bound | $\alpha$ | Upper bound |
|---|---|---|---|
| $(0,1)$ | $3.259708 \times 10^{-8}$ | $(1,0)$ | $3.258818 \times 10^{-8}$ |
| $(0,2)$ | $1.758240 \times 10^{-6}$ | $(1,1)$ | $1.466750 \times 10^{-6}$ |
| $(2,0)$ | $1.817997 \times 10^{-6}$ | $(0,3)$ | $1.534899 \times 10^{-4}$ |
| $(1,2)$ | $1.017039 \times 10^{-4}$ | $(2,1)$ | $1.004685 \times 10^{-4}$ |
| $(3,0)$ | $1.298543 \times 10^{-4}$ | | |

Combining these nine bounds with the mixed-derivative bounds through Lemma S3 gives all 34 first-, second- and third-order derivative bounds for $E = N - U$ on (S102). In the precise norm conventions of (S100), the outward-rounded budgets are

$$\begin{aligned} \Delta_0 &\le 7.561637 \times 10^{-8}, \quad \Delta_1 \le 1.102411 \times 10^{-5}, \\ \Delta_{\Sigma,1} &\le 2.242184 \times 10^{-5}, \quad \Delta_2 \le 2.483811 \times 10^{-3}. \end{aligned} \tag{S103}$$

The actual componentwise sums, rather than the potential-only dimension factors, are used. The zero value and first jet at the origin also give the sufficient uniform bound $\|E\|_{\infty,\mathcal{D}} \le 4.148040 \times 10^{-8}$ by Taylor's integral formula on this convex domain.

Separately, let $s_j$ denote the physical input scales and $D_j$ the declared normalized half-widths of the fitted architecture. On its real box $|x_j| \le s_j D_j$, a sufficient Legendre bound is $\widehat{H}_{pp} \succeq 0.55482152 I_2$. Here the positive finite-Gauss metric and an upper bound for $\|N_{pp}\|_2$ are combined as $\widehat{H}_{pp} \succeq (1/G_{\mathrm{upper}} - \|N_{pp}\|_2) I_2$. This real-box statement is distinct from the complex rectangle (S102); their momentum extents differ. It does not assert that $\widehat{H}_p = G^{-1}p$ or identify unobserved momentum dependence from rest labels.

Equations (S103) are finite learning-error outputs, not successful KAM smallness tests. A torus for this full-phase checkpoint requires its own residual, domain, geometry, response and torsion certificates. The structured model's previous invariant torus and executed magnetic-feedback holds are not transferred to it.

### *H.3 Power transport for a full-phase learned Hamiltonian*

This section concerns a candidate whose learned energy depends on both canonical position and momentum. It does not replace the prescribed-kinetic model certified in the article or transfer that model's controller certificate. All derivatives below use the same real canonical coordinates and clock. Let

$$H_*(q,p) = \tfrac{1}{2} p^{\mathsf{T}} G(q)^{-1} p + V_*(q), \qquad \widehat{H}(q,p) = H_*(q,p) + e(q,p),$$

where $G = SS^{\mathsf{T}} \succ 0$. Write

$$v_* = H_{*,p}, \quad \widehat{v} = \widehat{H}_p, \quad e_q = \widehat{H}_q - H_{*,q}, \quad e_p = \widehat{v} - v_*.$$

The two dissipative systems have damping matrices $R_* = SM_*S^\mathsf{T}$ and $\widehat{R} = S\widehat{M}S^\mathsf{T}$, respectively, with $\Delta R = \widehat{R} - R_*$. The learned factor $\widehat{M} = \mathrm{Re}(LL^*)$ is real symmetric positive semidefinite on real vectors. The magnetic port $g(q)u$ is shared.

*Learned-system identity.* For the equations

$$q' = \widehat{v}, \qquad p' = -\widehat{H}_q - \widehat{R}\widehat{v} + ug,$$

canonical cancellation gives

$$\widehat{H}' = -\widehat{v}^\mathsf{T}\widehat{R}\widehat{v} + ug^\mathsf{T}\widehat{v}.$$

Thus the PSD power sign remains valid for a full-phase energy. The Rayleigh derivative is taken with respect to an independent velocity in $\mathcal{D}(q, w) = w^\mathsf{T}\widehat{R}(q)w/2$, before setting $w = \widehat{H}_p$.

*Exact identity along the reference plant.* Along $q' = v_*$ and $p' = -H_{*,q} - R_*v_* + ug$, direct substitution gives

$$\begin{aligned}\widehat{H}' = & -\widehat{v}^\mathsf{T}\widehat{R}\widehat{v} + ug^\mathsf{T}\widehat{v} \\ & + e_q^\mathsf{T}\widehat{v} + e_p^\mathsf{T}(\widehat{R}\widehat{v} - \widehat{H}_q) + \widehat{v}^\mathsf{T}\Delta R v_*.\end{aligned}$$

Indeed, expansion first gives $-\widehat{v}^\mathsf{T}R_*\widehat{v} + ug^\mathsf{T}\widehat{v} + e_q^\mathsf{T}\widehat{v} + e_p^\mathsf{T}(R_*\widehat{v} - \widehat{H}_q)$. Replacing $R_* = \widehat{R} - \Delta R$ and $v_* = \widehat{v} - e_p$ proves the displayed identity.

Suppose sound pointwise or uniform certificates supply $|e_{q,j}| \le \varepsilon_{q,j}$, $|e_{p,j}| \le \varepsilon_{p,j}$ and $\|\widehat{M} - M_*\|_2 \le \delta_M$. Define

$$a^2 = \widehat{v}^\mathsf{T}G\widehat{v}, \qquad b^2 = \varepsilon_p^\mathsf{T}|G|\varepsilon_p,$$

where $|G|$ is entrywise absolute value. Then

$$\begin{aligned}\widehat{H}' \le & -\widehat{v}^\mathsf{T}\widehat{R}\widehat{v} + ug^\mathsf{T}\widehat{v} + \mathcal{E}, \\ \mathcal{E} = & \sum_j \varepsilon_{q,j}|\widehat{v}_j| + \sum_j \varepsilon_{p,j}|(\widehat{R}\widehat{v} - \widehat{H}_q)_j| + \delta_M a(a+b).\end{aligned}$$

To prove the last term, put $z = S^\mathsf{T}\widehat{v}$ and $w = S^\mathsf{T}v_*$. Then $|z^\mathsf{T}(\widehat{M} - M_*)w| \le \delta_M\|z\|_2\|w\|_2$ and $\|w\|_2 \le a + \|S^\mathsf{T}e_p\|_2 \le a + b$. The other two terms follow componentwise. If the certified kinetic energies coincide, $e_p = 0$, the expression reduces exactly to the existing potential-error budget plus $\delta_M v^\mathsf{T}Gv$.

The momentum-error term is necessary. In one degree of freedom take $H_* = (q^2 + p^2)/2$, $\widehat{H} = q^2/2 + p^2$, $R_* = \widehat{R} = 1$ and $u = 0$. Here $e_q = 0$ and $\Delta R = 0$, but the reference derivative is $-qp - 2p^2$, whereas the learned-system derivative is $-4p^2$. At $q = p \neq 0$ their difference is $p^2 > 0$. Consequently a potential-only error budget cannot certify this candidate.

*Energy and momentum-coordinate obligations.* On a convex domain containing zero, assume $e(0) = 0$, $De(0) = 0$ and $\|D^2e\|_2 \le \varepsilon_2$. Taylor's integral formula gives

$$|e(x)| \le \tfrac{1}{2}\varepsilon_2\|x\|_2^2.$$

If also $H_*(x) \ge m\|x\|_2^2/2$ and $0 \le \varepsilon_2 < m$, then

$$\widehat{H}(x) \ge \tfrac{1}{2}(m - \varepsilon_2)\|x\|_2^2, \qquad H_*(x) \le \frac{m}{m - \varepsilon_2}\widehat{H}(x).$$

This supplies an energy conversion only on that certified domain. Containment and forward-existence hypotheses remain necessary when using a sublevel set as an invariant region.

If a separately certified momentum-Hessian error is $\|e_{pp}\|_2 \le \varepsilon_{pp} < \inf \lambda_{\min}(G^{-1})$, then $\widehat{H}_{pp} \succ 0$. For each fixed $q$ it makes $p \mapsto \widehat{H}_p(q, p)$ strongly monotone on a convex momentum fiber and locally invertible. A prescribed velocity additionally has to lie in its image before asserting existence of a corresponding learned momentum. None of this permits replacing the observed physical momentum $p = G(q)v_*$ by a learned momentum without transforming the training balances.

For a continuous feedback, an inequality $-\widehat{v}^\mathsf{T}\widehat{R}\widehat{v} + ug^\mathsf{T}\widehat{v} + \mathcal{E} \le \kappa(c - \widehat{H})$ implies $\widehat{H} < c$, provided $\widehat{H}(x_0) < c$ and $\kappa > 0$ is finite, on any interval where the reference solution exists and remains in the verified domain. Indeed, $w = c - \widehat{H}$

satisfies $w' \geq -\kappa w$, and hence $w(t) \geq e^{-\kappa(t-t_0)} w(t_0) > 0$. A sampled controller additionally needs new entire-hold, execution and ledger certificates for that candidate. The algebra does not transfer the earlier 200+2 executed holds or their fixed budgets to a newly trained full-phase network.

## I Local and accumulated error of the conservative midpoint step

This proof supplies explicit constants for the accuracy statement in the article. Let $J_c = \left(\begin{smallmatrix} 0 & I_2 \\ -I_2 & 0 \end{smallmatrix}\right)$ and let $f = J_c \nabla \widehat{H}$ be $C^2$ on an open domain. Assume a common convex neighborhood contains all states and segments used below. Let $M, L, B_2$ bound $f, Df, D^2 f$, respectively, in a fixed vector norm and its induced multilinear norms. Exact trajectories and the unique local midpoint branches must remain in this domain through time $T$. These are domain hypotheses, not conclusions from a formal order calculation.

For an exact solution $x(t)$,

$$x'' = Df(x)f(x), \qquad x''' = D^2 f(x)[f(x), f(x)] + Df(x)Df(x)f(x).$$

Consequently $\|x''\| \leq LM$ and $\|x'''\| \leq M_3 := B_2 M^2 + L^2 M$. For $x_0 = x(0)$ and $x_h = x(h)$, the composite defect of the exact endpoint in the midpoint equation is

$$d_h = x_h - x_0 - hf((x_0 + x_h)/2).$$

The one-panel midpoint quadrature remainder for $\int_0^h f(x(t))\, dt$ is bounded by $M_3 h^3/24$. Moreover, linear interpolation of a twice differentiable trajectory gives

$$\|x(h/2) - (x_0 + x_h)/2\| \leq LMh^2/8.$$

Applying the Lipschitz constant $L$ to this chord-midpoint discrepancy yields

$$\|d_h\| \leq \left(\frac{M_3}{24} + \frac{L^2 M}{8}\right) h^3 = \left(\frac{B_2 M^2}{24} + \frac{L^2 M}{6}\right) h^3 = C_d h^3.$$

Both estimates follow by the integral Taylor remainder, so bounded continuous second derivatives of $f$ suffice.

Let $y = \Phi_h(x_0)$ be the exact solution of the implicit midpoint equation. Subtraction from the defect equation gives

$$\|y - x_h\| \leq (hL/2)\, \|y - x_h\| + \|d_h\|\,.$$

For $hL/2 \leq \kappa < 1$, the one-step bound is therefore $\|y - x_h\| \leq C_\ell h^3$ with $C_\ell = C_d/(1-\kappa)$.

Two initial states and their midpoint endpoints similarly satisfy

$$\|\Phi_h(x) - \Phi_h(y)\| \leq \frac{1 + hL/2}{1 - hL/2}\, \|x - y\| \leq e^{\Lambda h}\, \|x - y\|\,, \qquad \Lambda = L/(1-\kappa).$$

The last inequality follows by integrating $d \log[(1+a)/(1-a)]/da = 2/(1-a^2)$ and using $0 \leq a \leq \kappa$, or from $\log[(1+a)/(1-a)] \leq 2a/(1-a)$. Writing $e_n = \|x_n - x(nh)\|$ gives $e_{n+1} \leq e^{\Lambda h} e_n + C_\ell h^3$. For $e_0 = 0$,

$$e_n \leq C_\ell h^3 \frac{e^{\Lambda n h} - 1}{e^{\Lambda h} - 1} \leq C_\ell h^2 \frac{e^{\Lambda T} - 1}{\Lambda}.$$

Continuous limits cover $\Lambda = 0$. A nonzero initial error contributes the additional term $e^{\Lambda T} e_0$.

If an implementation returns $\widetilde{y}$ with algebraic residual $r = \widetilde{y} - x - hf((x + \widetilde{y})/2)$, contraction gives

$$\|\widetilde{y} - \Phi_h(x)\| \leq \frac{\|r\|}{1 - hL/2}.$$

These errors enter the same accumulated-error recursion. Per-step residuals of order $h^3$ preserve the stated global second-order rate under the same domain bounds. They do not imply that the computed map satisfies the exact symplectic Jacobian identity.

## References


Abbott JJ, Diller E and Petruska AJ (2020) Magnetic methods in robotics. *Annual Review of Control, Robotics, and Autonomous Systems* 3: 57–90. doi: 10.1146/annurev-control-081219-082713

Antman SS (2005) *Nonlinear Problems of Elasticity*. 2nd ed. Springer. doi: 10.1007/0-387-27649-1

Chen Y, Matsubara T and Yaguchi T (2022) KAM theory meets statistical learning theory: Hamiltonian neural networks with non-zero training loss. *Proceedings of the AAAI Conference on Artificial Intelligence* 36(6): 6322–6332. doi: 10.1609/aaai.v36i6.20582

Desai SA, Mattheakis M, Sondak D, Protopapas P and Roberts SJ (2021) Port-Hamiltonian neural networks for learning explicit time-dependent dynamical systems. *Physical Review E* 104: 034312. doi: 10.1103/PhysRevE.104.034312

Greydanus S, Dzamba M and Yosinski J (2019) Hamiltonian neural networks. In: *Advances in Neural Information Processing Systems*, vol. 32. https://arxiv.org/abs/1906.01563.

Hairer E (2010) *Geometric Numerical Integration*, Lecture 2: Symplectic integrators. TU München, January–February 2010, Theorem 3, pp. 2–3. https://www.unige.ch/~hairer/poly_geoint/week2.pdf.

Haro À and Luque A (2019) A-posteriori KAM theory with optimal estimates for partially integrable systems. *Journal of Differential Equations* 266(2–3): 1605–1674. doi: 10.1016/j.jde.2018.08.003 Preprint version used: https://arxiv.org/abs/1711.05515v1.

Kim Y, Parada GA, Liu S and Zhao X (2019) Ferromagnetic soft continuum robots. *Science Robotics* 4(33): eaax7329. doi: 10.1126/scirobotics.aax7329

Pöschel J (2001) A lecture on the classical KAM theorem. In: *Proceedings of Symposia in Pure Mathematics*, vol. 69, pp. 707–732. doi: 10.1090/pspum/069/1858551 Preprint version used: https://arxiv.org/abs/0908.2234v1 (Version 1.3, 2009).

Simo JC (1985) A finite strain beam formulation. The three-dimensional dynamic problem. Part I. *Computer Methods in Applied Mechanics and Engineering* 49(1): 55–70. doi: 10.1016/0045-7825(85)90050-7

Trabelsi C, Bilaniuk O, Zhang Y et al. (2018) Deep complex networks. In: *International Conference on Learning Representations*. https://arxiv.org/abs/1705.09792.

Zhang SQ, Gao W and Zhou ZH (2022) Towards understanding theoretical advantages of complex-reaction networks. *Neural Networks* 151: 80–93. doi: 10.1016/j.neunet.2022.03.024 Published version used for theorem numbering.

Zhong YD, Dey B and Chakraborty A (2020) Dissipative SymODEN: Encoding Hamiltonian dynamics with dissipation and control into deep learning. In: *ICLR Workshop on Integration of Deep Neural Models and Differential Equations*. https://arxiv.org/abs/2002.08860.